ARCADIA MINICA

OFFICIAL NO. MAR-2400-D

YARD NO.: 113

TRIM & STABILITY BOOKLET

DRG NO: EEPL/113/20

OWNER

ARVIND & CO. SHIPPING AGENCIES PVT. LTD.
CITY POINT, 5TH FLOOR, OPP. TOWN HALL
JAMNAGAR, GUJARAT 361001

BUILDER

ESSFOUR ENGINEERING PVT. LTD,
A4 GATEWAY COMMUNE,
CANAL ROAD, VESU,
GUJARAT - SURAT

AS AMENDED
ON PAGE NO:3,4,5,8,10,16,27

TOTAL 51 SHEETS

I N D E X

GENERAL PARTICULARS

1.	Vessel's Name	:	ARCADIA MINICA
2.	Official Number	:	MAR-2400-D
3.	Port of Registry	:	MORMUGAO
4.	Owner's Name and Address	:	Arvind & Co. Shipping Agencies Pvt Ltd City Point, 5th Floor, Opp. Town Hall Jamnagar, Gujarat 361001
5.	Builder's Name and Address	:	ESSFOUR ENGINEERING (P) LTD, A4 GATEWAY COMMUNE, CANAL ROAD, VESU GUJARAT - SURAT
6.	Yard No.	:	113
7.	Year of Built	:	2022
8.	Principal Dimensions		
	a. Length O.A.	:	55.00 m.
	b. Breadth moulded	:	16.00 m.
	c. Depth moulded	:	03.00 m.
	d. Draft (loaded)	:	02.20 m.
9.	Displacement (Loaded)	:	1765.520 tonnes
10.	Deadweight	:	1376.885 tonnes
11.	Gross Tonnage	:	646
12.	Net Tonnage	:	194
13.	Class Notation	:	卐 SU, PONTOON, Indian Costal Services, Deck Strengthemed for 9T/sqm, for Crane Operation
14.	Sign Convension	:	Longitudinal Position (LCG, LCB, LCF etc) are measured from Midship Fr. No. 15 (27mtrs from Fr No.0) Aft of AP is considered –ve and forward +ve. All vertical measurements (VCG, VCB etc) are measured from baseline. TCG are measured wrt Centerline Port –ve Stbd +ve. Trim Aft + ve Trim Ford - ve

NOTE:
Deck loading shall not exceed 15 t/m2 up to 4.0 m P & S and 9 t/m2 elsewhere(Refer "Scantling Section").Accordingly, Operator has to ensure that pressure on deck due to crane / cargo loads should not exceed the same on main deck structure.

2. Notes on Stability

2.1 Introduction

Compliance with the required minimum stability criteria does not ensure immunity against capsizing regardless of the circumstances, or absolve the Master for his responsibilities. Master should therefore exercise the prudence and good seamanship having the regard to the season of the year, weather forecasts, the navigational zone, and should take the appropriate action as to speed and course warranted by prevailing circumstance.

2.2 Operational Limits

The following operational limits should be noted:

- Maximum draught shall be as per issued Load Line Certificate.
- ~~The Value of VCG during the actual loading conditions corrected by the free surface moment must always be within the safe range of maximum allowable VCG curve. The vessel should not be operated at a trim exceeding the values of the trimmed maximum allowable VCG curves.~~

2.3 Closing of opening

Flooding is a constant source of danger to the safe and efficient operation of any ship, and may cause the contamination of fuel, loss of electrical power, loss of engine power and damage of cargo. In addition, the flooding may cause loss of cargo, loss of buoyancy and loss of stability, which may further cause the serious listing and sometimes lead to capsizing and the total loss of the ship.

Flooding can be dangerous, and therefore there is a need to always be vigilant against its occurrence. In particular, it is essential to ensure that good seamanship is always exercised. As a consequence, it is strongly recommended that the operational procedures listed below should be strictly adopted, whenever appropriate, on an individual ship basis.

Before Departure & During voyage

Ensure that:

All manholes serving spaces below the main deck are to be effectively closed and secured watertight, or weathertight where applicable, before the ship leaves port and to be kept closed during navigation. The times of opening and closing of the applicable closing devices are to be recorded in the official log as required by the regulations.

2.4 Stability and Freeboard during loading and unloading

It is required to comply with the stability and freeboard requirement during the process of loading and unloading. The ship is then in condition to carry out these operations safely. The Master should pay special attention when loading or unloading heavy liquids/weights, in order to avoid excessive asymmetrical heeling moments. In particular, the Master shall ensure that:

- The ship has adequate stability
- The freeboard at any door/openings giving access to the hull is sufficiently closed to prevent the entry of water
- Cargo should be unloaded from the top (heaviest items first).

2.5 Operational Precautions against Capsizing

This booklet is made for the guidance of the master in order to ensure safe operation and proper loading of the vessel.

The vessel is assumed to be floating in salt water of sp gr 1.025.

The Master of the vessel should read the entire booklet and be familiar with the characteristics of the vessel before placing it into service. Any ignorance of the instructions in this booklet may jeopardize the safety of crew and vessel.

The loading conditions shown in this booklet are typical for the intended service of the vessel and are intended as guidance. It is thus emphasized that a separated calculation is necessary for all actual loading conditions. Care should be taken to ensure that the cargo allocated to the ship is capable of being stowed so that compliance with the required stability criteria can be achieved.

Before a voyage commences, care should be taken to ensure that the cargo and sizable pieces of equipment have been properly stowed and secured/lashed so as to minimize the possibility of both longitudinal and lateral shifting, while at sea under the effect of acceleration caused by rolling and pitching.

The Master should note that the stability can be adversely affect by such influences as beam wind on ship with large wind area and deck cargo, water trapped on deck and in deck cargo area, rolling characteristics and following seas. In severe weather, the speed of the ship should be reduced if excessive rolling, propeller emergence, shipping of water on deck or heavy slamming occurs. Six heavy slamming or 25 propeller emergences during 100 pitching motions should be considered dangerous.

Special attention should also be paid when a ship sailing in following or quartering sea because dangerous phenomena such as parametric resonance, broaching to, reduction of stability on the wave crest, and excessive rolling may occur singularly, in sequence or simultaneously in a multiple combination, creating a threat of capsize. Particularly dangerous is the situation when the wave length is of the order of 1.0 to 1.5 ship's length. A ships speed and/or course should be altered appropriately to avoid the above-mentioned phenomena.

Care should be taken to the limit of the amount and height of the deck cargo so that the compliance with the stability criteria can be achieved. When the deck cargo is intended to be carried on a fully load departure condition at the maximum summer draft, the vessel will at no point in time have a KG exceeding the limiting KG curve.

No modification or removal can be allowed for air pipes, ventilators and other piping systems without authorization of the classification society.

The bending moment and shear force of each loading condition must not exceed the permissible values indicated by the classification society.

2.6 Intact Stability Criteria

2.6.1. The area under the righting lever curve up to the angle of maximum righting lever should not be less than 0.08 metre-radians

* 2.6.2. The static angle of heel due to uniformly distributed wind load of 540Pa (wind speed 30m/s) should not exceed an angle corresponding to half the freeboard for the relevant loading condition, where the lever of wind heeling moment is measured from the centroid of the windage area to half the draught.

2.6.3. The minimum range of stability should be 20°($L \leq 100$ m)

\# 2.6.4. Lifting:

The residual righting area below the righting lever and above the heeling lever curve between φe and the lesser of 40° or the angle of the maximum residual righting lever should not be less than 0.080 m rad.

The equilibrium angle is to be limited to the lesser of the following:

A. 10 degrees;

B. The angle of immersion of the highest continuous deck enclosing the watertight hull;

C. The lifting appliance allowable value of trim/heel (data to be derived from side lead and off lead allowable values obtained from manufacturer)

2.7 Crane Operation

One crawler crane would be carried and operated on the pontoon, a large heeling moment can be applied to the vessel when the crane is in use. It is the responsibility of the Master to control crane operations to ensure that the resultant heel angle of the ship does not exceed allowable limit.

THE OPERATION POSITION OF CRANE WAS LOCATED ON THE CENTRE OF VESSEL. (27.5m from Fr 0, CENTRELINE)

MAXIMUM LOAD OF THE CRANE DURING OPERATION SHOULD BE LIMITED TO 30T

MAXIMUM WORK RADIUS OF THE CRANE DURING OPERATION SHOULD BE LIMITED TO 21.45m (i.e. crane outreach should not exceed 13.45m outboard of the ship sides).

\# Intact stability during crane operations has been independently verified

Calculations are shown at Section 13.4. Crane dimensions are shown in Fig 2.7.1.

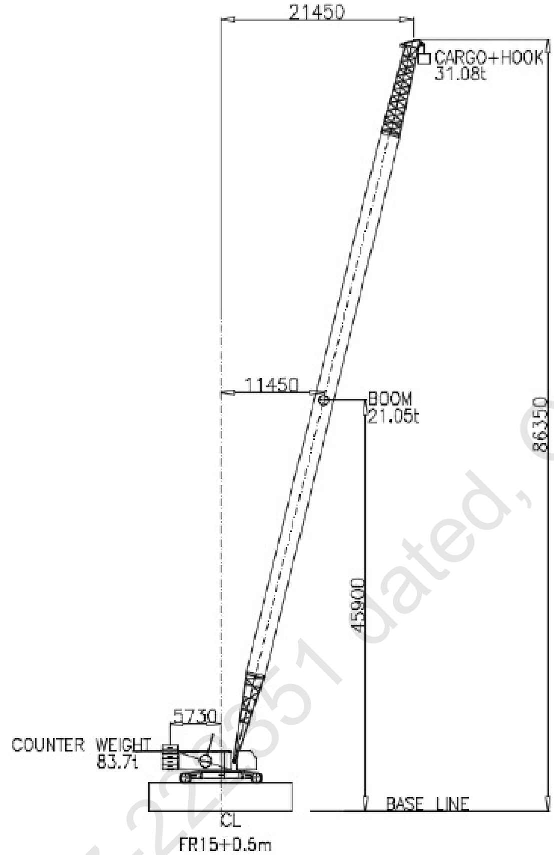


Fig 2.7.1

ITEM	Weight	Ycg	Zcg
	t	m, from CL	m, from BL
CRANE BOOM	21.05	11.45	45.900
CARGO+HOOK	31.08	21.45	86.350
COUNTER WEIGHT	83.7	-5.73	5.730

CRANE LOAD BREAKUP

Item	Weight	LCG	Lmom	VCG	Vmom	TCG	FSM
CRANE	126.5	0.5	63.25	4.5	569.25	0	0
CRANE BOOM	21.1	0.5	10.55	45.9	968.49	0	0
COUNTER WEIGHT	83.7	0.5	41.85	5.73	479.6	0	0
TOTAL	231.3	0.5	115.65	8.721747	2017.34	0	0

METRIC CONVERSIONS TABLE

METRIC EQUIVALENTS

THE USE OF S.I (SYSTEM INTERNATIONALS) UNITS IS STRONGLY RECOMMENDED.

MULTIPLY BY BELOW	TO CONVERT FROM	TO OBTAIN	-
0.03937	Millimetres	Inches	25.400
0.3937	Centimetres	Inches	2.5400
3.2808	Metres	Feet	0.3048
2.2046	Kilogrammes	Pounds	0.45359
0.0009842	Kilogrammes	Tons (2240lbs)	1016.047
0.9842	Metric Tons (i.e.Tonnes of 1000 Kilos)	Tons (2240lbs)	1.016
2.4998	Metric Tons Per Centimetre Of Immersion	Tons Per Inch (Immersion)	0.4000
8.2014	Moment To Change Trim One Centimetre (Tonne Metre Units)	Moment To Change Trim One Inch (Foot Ton Unit)	0.122
187.9767	Metre Radians	Feet Degrees	0.0053
-	TO OBTAIN	TO CONVERT FROM	MULTIPLY BY ABOVE

RELATION BETWEEN WEIGHT AND VOLUME

1000 cubic milliimeter	=	1 cubic centimetre
1 cubic metre	=	35.316 Cubic feet
1 cubic foot	=	0.0283 Cubic metres
1 cubic centimeter of freshwater (S.G. 1.0)	=	1 gramme
1000 cubic centimeters of freshwaters (S.G. 1.0)	=	1 kilogram (1000 grammes)
1 cubic metre of freshwater (S.G. 1.0)	=	1 Tonne (1000 Kilos)
1 cubic metre of saltwater(S.G.1.025)	=	1.025 Tonnes
1 tonne of saltwater (S.G. 1.025)	=	0.975 Cubic metres

NOTES ON USE OF FREE SURFACE MOMENTS

Provided a tank is completely filled with liquid, no movement of the liquid is possible and the effect on the ship's stability is precisely the same as if the tank contained solid materials.

Immediately a quantity of liquid is withdrawn from the tank, the situation changes completely and the stability of the ship is adversely affected by what is known as the " free surface effect". This adverse effect on the stability is referred to as a " loss in G.M." or as a " virtual rise in V.C.G." and is calculated as follows:

$$\text{Loss in G.M. due to Free Surface Effects (in metres)} = \frac{\text{Free Surface Moment (tonne - metre)}}{\text{Displacement of vessel (tonnes)}}$$

N. B.: The "Free Surface Effects" of a proportion of all oil, fuel, fresh water and service tanks should be taken into account in both the Arrival and Departure Conditions.

$$\begin{matrix}\text{Free Surface Moment} \\ \text{(tonnes- metre)} \\ \text{of the liquid}\end{matrix} = \left\{\begin{matrix}\text{Free Surface Moment} \\ \text{at Sp. Gr. 1.0} \\ \text{(tonnes – metre)}\end{matrix}\right\} \times \left\{\begin{matrix}\text{Sp. Gr. Of} \\ \text{liquid}\end{matrix}\right\}$$

NOTE:

1. ~~In any condition where ballast has been taken, care must be taken to ensure that the tanks are fully pressed up to eliminate unnecessary Free Surface Effects.~~
2. ~~Bunkers should be drawn from one respective tank so that other tanks are maintained in the pressed up condition.~~
3. ~~Slack must not be kept in all tanks. It must be collected in one tank.~~

WORKED OUT EXAMPLE OF STABILITY CALCULATION

Calculations in this Booklet are preformed by using program Hydrostatics & stability program suite Version 06.07.05 fv 16.6, Developed by Wolfson unit m.t.i.a Southampton university, England. Below mentioned loading sheet provides a simplified means of determining available GM and draft at various conditions of vessel loading. This sheet is largely self explanatory. After totaling various items of vessel loading as indicated by the loading tables shown, the vessel's draft and available (corrected) GM may be determined in a simple, direct and accurate manner.

LOADING SHEET

LOAD DEPATURE CONDITION WITH CRANE STOWAGE

A	B	C	D	E	F	G
LOCATION/ COMPARTMENT/TANKS	WEIGHT IN TONNES	LCG AT FR. No. 15 IN MTRS	LONGL Moment IN T-M	VCG@BASE IN MTRS	VERTL Moment IN T-M	F.S.M IN T-M
CRANE	231.250	0.500	115.63	4.930	1140.063	0.000
LIGHTSHIP	388.635	1.332	517.66	3.000	1165.905	0.000
	Σ Weights	D ÷ B	Σ LONGL. Mom.	F ÷ B	Σ VERTL. Mom.	Σ' F.S.M
	619.885	**1.022**	**633.29**	**3.720**	**2305.968**	**0.000**

FROM HYDROSTATICS REPORT FOR CORRESPONDING DISPLACEMENT

DRAFT AT MIDSHIP	0.853	Mtrs.	TRIM LEVER = LCG - LCB	-0.125	Mtrs.
LCB FROM MIDSHIP	0.897	Mtrs.	TRIM MOMENT = DISPL. X TRIM LEVER	-77.250	T- M
LCF FROM MIDSHIP	0.816	Mtrs.			
LCG FROM MIDSHIP	1.022	Mtrs.	TRIM BY AFT/FORD ON DARFT MARKS=TRIM MOMENT / (MCTIcm x 100)	-0.019	Mtrs.
TPC	7.761	T/cm			
MCT1CM	40.272	T- M			
KMT	27.365	Mtrs.	TRIM AFT AT FR 05 = {(18+ LCF) x TRIM }/36	-0.010	Mtrs.
VCG	3.720	Mtrs.			
GMT Solid = KMT - VCG	23.645	Mtrs.	TRIM FORD AT FR 25 = {(18 - LCF) x TRIM AFT / 36	-0.009	Mtrs.
F.S. Corrn. = Σ F.S.M ÷ Σ Weights	0.000	Mtrs.	DRAFT AFT AT FR 05 = DRAFT LCF + TRIM AFT	0.843	Mtrs.
G.M.T Fluid = G .M.T Solid - F.S. Corrn.	23.645	Mtrs.	DRAFT FORD AT FR 25= DRAFT LCF - TRIM FORD	0.862	Mtrs.

INSTRUCTION FOR USE OF LOADING SHEET

- Enter the Deadweight in Tonnes of cargo, liquids, Stores and crew carried in the various locations, compartments and tanks in colomn A. This breaks down the loading of the ship into the appropriate Zones.

- Enter Corresponding LCG & VCG W.R.T Fr. No. 15 in column C & E respectively. Aft of Fr No. 15 is –ve and Ford of Fr. No. 15 is +ve

- Calculated Vertical Moments (ie Weight x VCG) & Longl. Moment (Weight x LCG)

- Add the entries in column B,D, F & G to obtain the DISPLACEMENT / Vertical Moment / Longl. Moment/F.S.Moment
  Now that the Displacement is established, the draft & hydrostatic data from Hydrostatic Report for corresponding Displacement can be entered.

- GMT solid, GMT fliud, Trim Lever, Trim Moment, Trim by aft/ford and draught aft & ford to calculate as shown in the loading sheet

- The tabular GM in Loading sheet includes an allowance for slack tank free surface for each type of consumable liquid.

A GM available less than the GM required indicates insufficient stability. ~~Such a condition may be corrected by ballasting tanks in any zone having a positive stability factor.~~

DEFINITIONS, SYMBOLS AND SIGN CONVENTIONS.

Midship at	Fr. No. 15
Frame Spacing	1800 mm from Fr. 0 to Fore end

List of Symbols

Symbol	**Unit**	**Description**
LOA	M	Length Overall
LWL	M	Length Waterline
B (Mld)	M	Moulded Breadth
D (Mld)	M	Moulded Depth
TPCm	T	Tonnes per 1 Cm immersion
MCTCm	T-M	Moment to change Trim by 1 Cm
Draught at ~~Fr. 21~~Aft Mark	M	Draught at Fr. 05
Draught Mid	M	Draught from Base line at Midship
Draught at ~~Fr 48~~Fwd mark	M	Draught at Fr. 25
VCG, KG	M	Vertical Center of Gravity above Base line before correcting for free surface.
KGc	M	Vertical Center of Gravity above Base line after correcting for free surface.
KB	M	Vertical Center of Buoyancy above Base line
KMt	M	Transverse Metacenter above Base line
GoMt (Fluid)	M	Transverse Metacenter above KGc
GZ	M	Righting Arm
FSM	T-M	Free Surface Moment
Sp.Gr.		Specific Gravity

EXAMPLE SHOWING USE OF CROSS CURVES

The Purpose of Cross Cruves of Stability is to enble the Statical Stability cruves to be drawn for the ship in the sailing condition.

For example assuming the displacement of the ship to be 619.885 tonnes

The corrected Vertical Centre of Gravity "KGc" above Base line to be 3.720 mtrs

Procedure to obtain 'GZ' levers is as follows:-

Obtain 'KN' values for the displacement of 619.885 Tonnes for various angle of Heel from 0° to 80° at 10° interval.

Righting Lever GZ = KN - KGcSin θ

Where KN is Cross Curve ordinates in metres

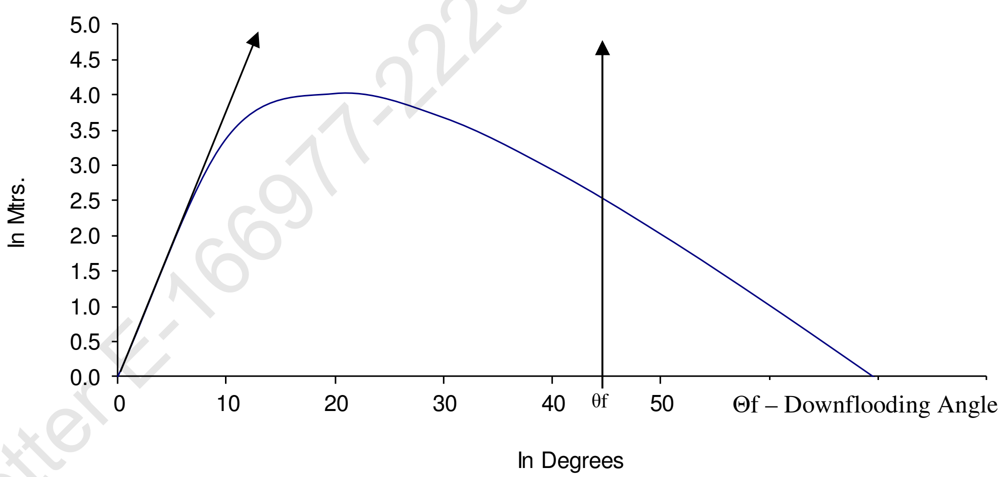
KGc is Centre of Gravity above base line corrected for the free surface effect in metres

And 'θ' is Angle of Heel in degrees

By Plotting GZ levers against angle of heel 'θ' the statical stability curves for the ship for any displacement can be plotted.

The table below illustrates the above calculation

IN DEGREES	0	10	20	30	40	50	60	70	80
SIN Θ	0.000	0.174	0.342	0.500	0.643	0.766	0.866	0.940	0.985
KG SIN Θ	0.000	0.646	1.272	1.860	2.391	2.850	3.222	3.496	3.663
KN	0.000	4.037	5.301	5.548	5.339	4.883	4.244	3.46	2.564
GZ = KN-KGc SINΘ	0.000	3.391	4.029	3.688	2.948	2.033	1.022	-0.036	-1.099



STATUTORY REQUIREMENTS:

No.	Criteria	Requirement	Pass/Fail
1	The area under the GZ curve from 0.0 degrees to 20.5 degrees (Max. GZ)	0.08 m.rad	pass
2	The vanishing stability angle	20 degrees	pass
3	The heel due to applied heeling curve	8.6 degrees	pass

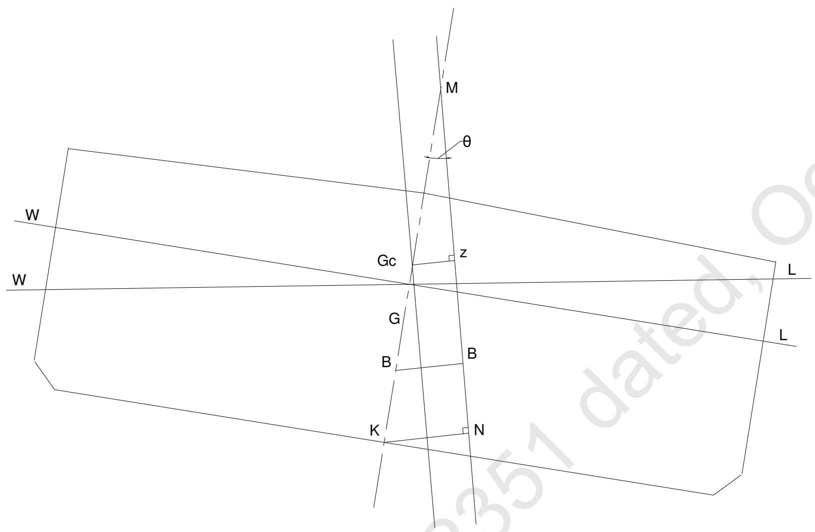
USE OF CROSS CURVES OF STABILITY

The Cross curves of stability have been calculated for centre of gravity w.r.t to baseline. In any condition, Where the centre of gravity is KG and free Surface Effects GGc, the righting lever.

G Z = KN - KGc Sin θ, Where

KGc = KG + GGc

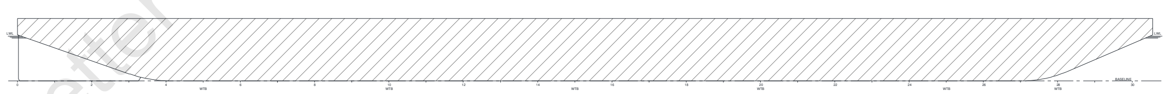
No Erections included in the cross curves



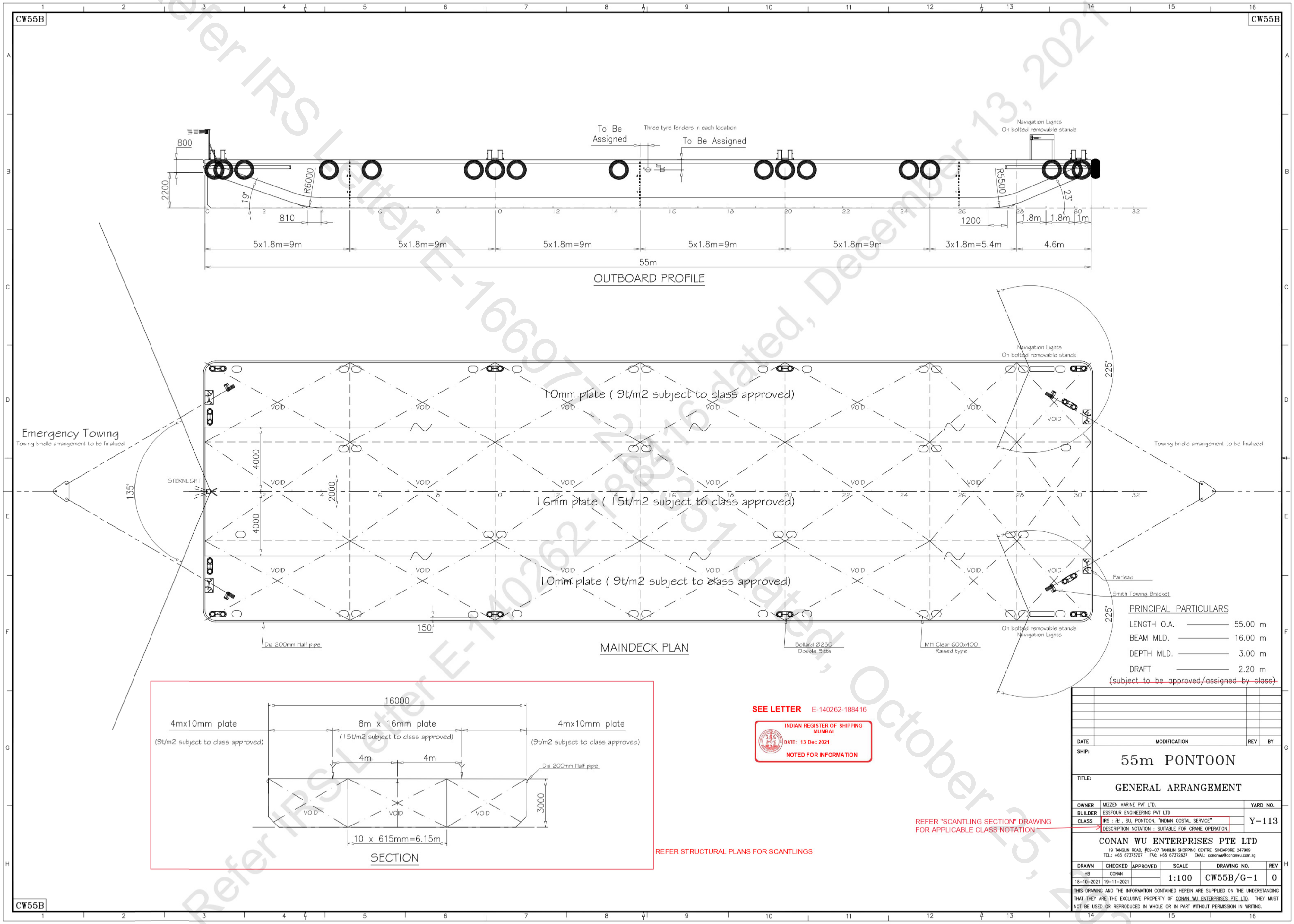
Note : -

1. KN is obtained from the Cross Curves for corresponding Displacement in tonnes.
2. KGc is vertical centre of gravity in the condition computed.

VOLUME USE FOR KN COMPUTATIONS



Frame Spacing
From Fr. 0 to Forepeak= 1800 mm


OUTBOARD PROFILE
To Be Assigned
Three tyre fenders in each location
To Be Assigned
Navigation Lights
On bolted removable stands
800
2200
19°
R6000
810
R5500
23°
1200
1.8m
1.8m
1m
5x1.8m=9m
5x1.8m=9m
5x1.8m=9m
5x1.8m=9m
5x1.8m=9m
3x1.8m=5.4m
4.6m
55m
MAINDECK PLAN
Emergency Towing
Towing bridle arrangement to be finalized
STERNLIGHT
135°
225°
225°
10mm plate (9t/m2 subject to class approved)
16mm plate (15t/m2 subject to class approved)
10mm plate (9t/m2 subject to class approved)
VOID
4000
4000
2000
150
Dia 200mm Half pipe
Bollard Ø250
Double Bitts
MH Clear 600x400
Raised type
Navigation Lights
On bolted removable stands
Fairlead
Smith Towing Bracket
Towing bridle arrangement to be finalized
PRINCIPAL PARTICULARS
LENGTH O.A. 55.00 m
BEAM MLD. 16.00 m
DEPTH MLD. 3.00 m
DRAFT 2.20 m
(subject to be approved/assigned by class)
SECTION
16000
4mx10mm plate
(9t/m2 subject to class approved)
8m x 16mm plate
(15t/m2 subject to class approved)
4mx10mm plate
(9t/m2 subject to class approved)
4m
4m
Dia 200mm Half pipe
3000
10 x 615mm=6.15m
REFER STRUCTURAL PLANS FOR SCANTLINGS
SEE LETTER E-140262-188416
INDIAN REGISTER OF SHIPPING
MUMBAI
DATE: 13 Dec 2021
NOTED FOR INFORMATION
REFER "SCANTLING SECTION" DRAWING
FOR APPLICABLE CLASS NOTATION
DATE
MODIFICATION
REV
BY
SHIP: 55m PONTOON
TITLE: GENERAL ARRANGEMENT
OWNER MIZZEN MARINE PVT LTD.
BUILDER ESSFOUR ENGINEERING PVT LTD
CLASS IRS : ✠, SU, PONTOON, "INDIAN COSTAL SERVICE"
DESCRIPTION NOTATION : SUITABLE FOR CRANE OPERATION.
YARD NO. Y-113
CONAN WU ENTERPRISES PTE LTD
19 TANGLIN ROAD, #09-07 TANGLIN SHOPPING CENTRE, SINGAPORE 247909
TEL: +65 67373707 FAX: +65 67372637 EMAIL: conanwu@conanwu.com.sg
DRAWN HB 18-10-2021
CHECKED CONAN 19-11-2021
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REV 0
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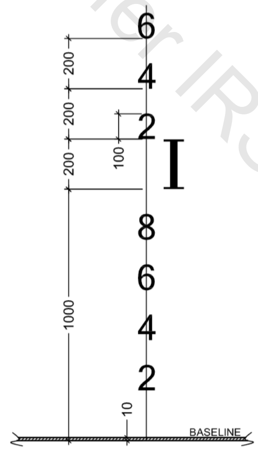
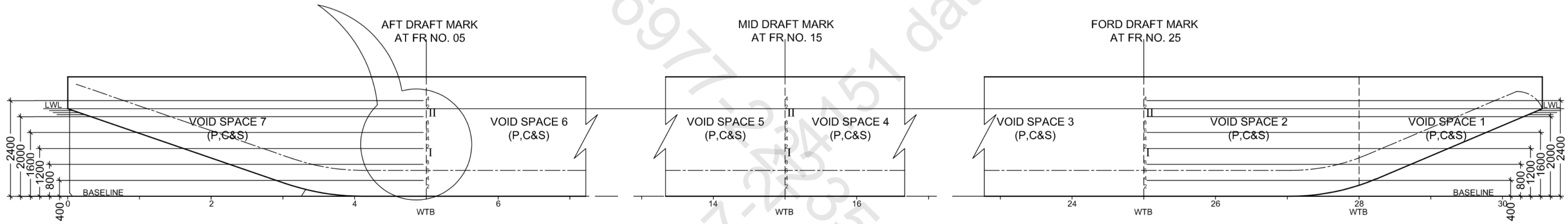
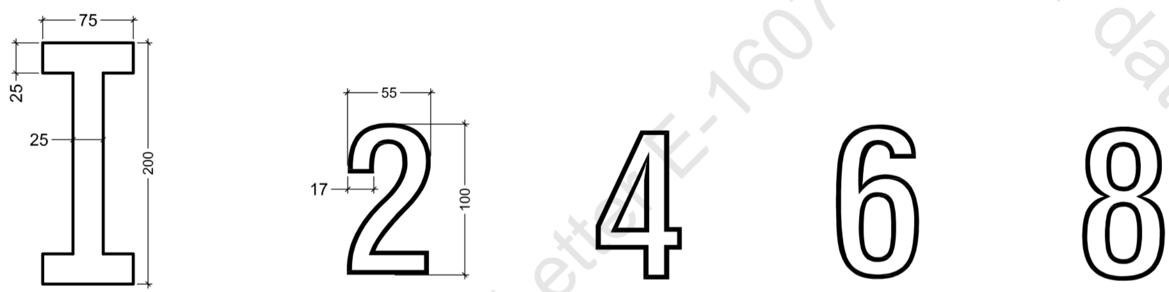


TYPICAL DRAFT MARK



PROFILE



MAIN DIMENSIONS

LENGTH	O.A	55.00 M
BREADTH	MLD	16.00 M
DEPTH	MLD	03.00 M
DRAUGHT	MLD	02.20 M

NOTES ON DRAUGHT MARKS
1) AFT, MID & FORD DRAUGHT MARKS TO BE MARKED W.R.T. UNDERSIDE OF KEEL
2) DRAUGHT MARKS TO BE PAINTED WHITE
3) HEIGHT OF DRAUGHTS MARKS NUMBERS TO BE PROJECTED, IF NOT PERPENDICULAR TO BASE LINE AT SITE. BOTTOM OF THE NUMBER INDICATES THE WATER LEVEL FROM U.S.K, TOP OF NUMBER SHOULD BE 100mm. ABOVE BOTTOM OF NUMBER WHEN BE MEASURED PERPENDICULAR TO BASE LINE & (200mm. FOR ROMAN NUMBER)
4) DRAUGHT MARK TO BE MARKED ON PORT & STARBOARD
5) DRAUGHT MARKS TO BE FABRICATED FROM 8mm. PLATE TO BE FULLY WELDED & PAINTED WHITE

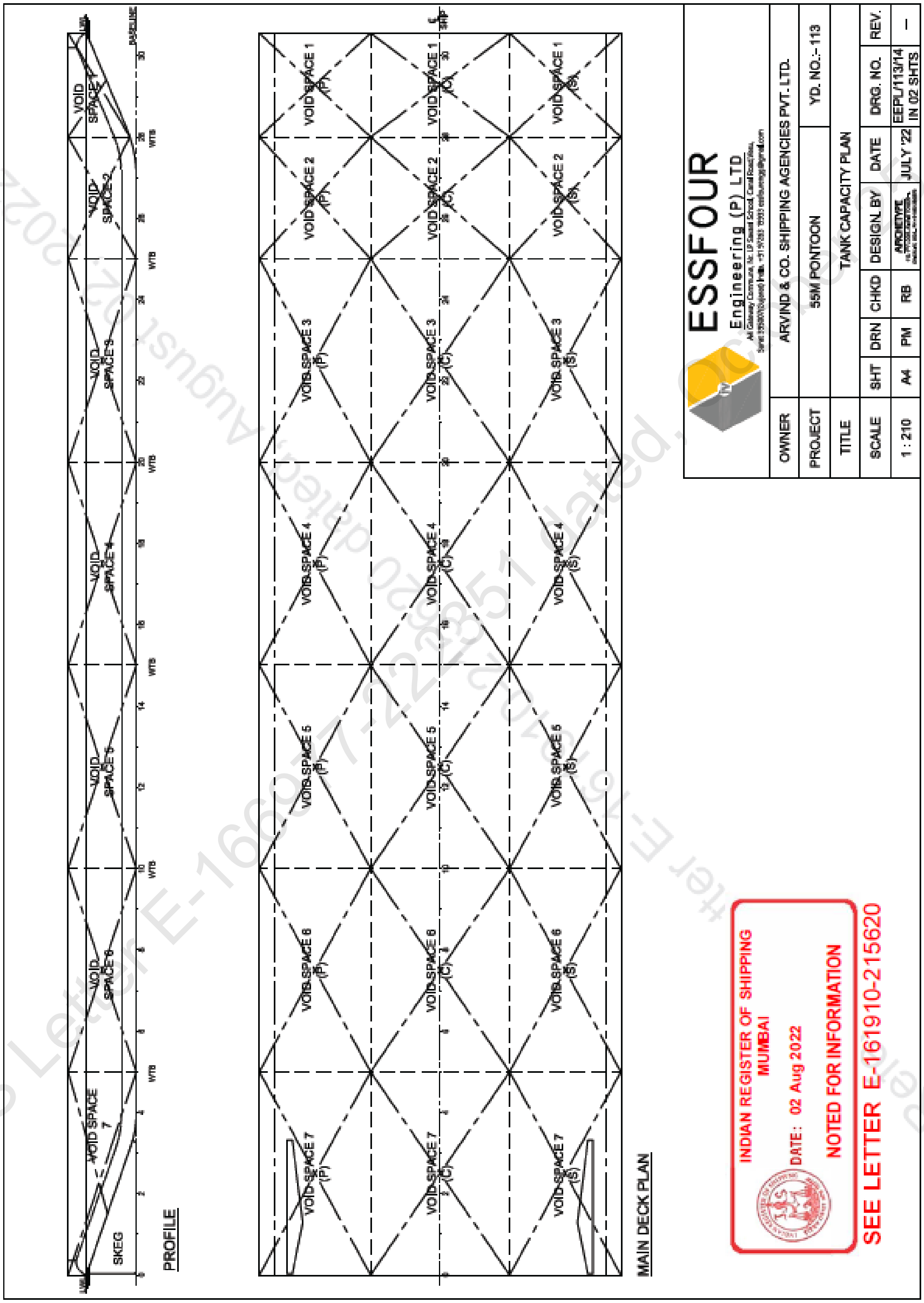
INDIAN REGISTER OF SHIPPING
MUMBAI
DATE: 04 Aug 2022
NOTED FOR INFORMATION

SEE LETTER E-160767-214151

ESSFOUR
Engineering (P) LTD
A4 Gateway Commune, Nr. LP Savani School, Canal Road,Vesu, Surat 395007(Gujarat) India, +91 97263 19933 essfourengg@gmail.com

OWNER	ARVIND & CO. SHIPPING AGENCIES PVT. LTD.						
PROJECT	55M PONTOON					YD. NO.:- 113	
TITLE	DRAUGHT MARK PLAN						
SCALE	SHT	DRN	CHKD	PREP. BY	DATE	DRG. NO.	REV.
1 : 100	A3	CG	RB	ARCHETYPE	JUL. '22	EEPL/113/02	---

TANK CAPACITY PLAN



SR. NO.	DESCRIPTION	LOCATION OF FR. NO.	VOLUME (CU.M)	DENSITY (T/M^3)	CAPACITY (TONNES)	L.C.G WRT MIDSHIP (M)	V.C.G. WRT B.L. (M)	T.C.G.WRT C.L. SHIP (M)
01	VOID SPACE 7 (C)	FR NO. 00 - 05	122.5	0.001	0.123	-21.666	1.770	0
02	VOID SPACE 7 (P)	FR NO. 00 - 05	96.0	0.001	0.096	-21.459	1.770	-5.493
03	VOID SPACE 7 (S)	FR NO. 00 - 05	96.0	0.001	0.096	-21.459	1.770	5.493
04	VOID SPACE 6 (C)	FR NO. 05 - 10	166.1	0.001	0.166	-13.500	1.500	0
05	VOID SPACE 6 (P)	FR NO. 05 - 10	131.1	0.001	0.131	-13.500	1.500	-5.538
06	VOID SPACE 6 (S)	FR NO. 05 - 10	131.1	0.001	0.131	-13.500	1.500	5.538
07	VOID SPACE 5 (C)	FR NO. 10 - 15	166.1	0.001	0.166	-4.500	1.500	0
08	VOID SPACE 5 (P)	FR NO. 10 - 15	131.1	0.001	0.131	-4.500	1.500	-5.538
09	VOID SPACE 5 (S)	FR NO. 10 - 15	131.1	0.001	0.131	-4.500	1.500	5.538
10	VOID SPACE 4 (C)	FR NO. 15 - 20	166.1	0.001	0.166	4.500	1.500	0
11	VOID SPACE 4 (P)	FR NO. 15 - 20	131.1	0.001	0.131	4.500	1.500	-5.538
12	VOID SPACE 4 (S)	FR NO. 15 - 20	131.1	0.001	0.131	4.500	1.500	5.538
13	VOID SPACE 3 (C)	FR NO. 20 - 25	166.1	0.001	0.166	13.500	1.500	0
14	VOID SPACE 3 (P)	FR NO. 20 - 25	131.1	0.001	0.131	13.500	1.500	-5.538
15	VOID SPACE 3 (S)	FR NO. 20 - 25	131.1	0.001	0.131	13.500	1.500	5.538
16	VOID SPACE 2 (C)	FR NO. 25 - 28	98.6	0.001	0.099	20.678	1.514	0
17	VOID SPACE 2 (P)	FR NO. 25 - 28	77.8	0.001	0.078	20.678	1.532	-5.505
18	VOID SPACE 2 (S)	FR NO. 25 - 28	77.8	0.001	0.078	20.678	1.532	5.505
19	VOID SPACE 1 (C)	FR NO. 28 - 30+	50.1	0.001	0.050	25.281	2.026	0
20	VOID SPACE 1 (P)	FR NO. 28 - 30+	39.3	0.001	0.039	25.281	2.036	-5.490
21	VOID SPACE 1 (S)	FR NO. 28 - 30+	39.3	0.001	0.039	25.281	2.036	5.490

NOTE :
1. MIDSHIP IS 27M FWD OF TRANSOM (FR. NO. 0).

Hydrostatics Report

55M Pontoon

Ship Particulars

Waterline Length	55.000 metres
Waterline Beam	16.000 metres
Top of Keel	0.000 metres
CP and CM referred to Section	55M Pontoon, FR NO. 15: X=0 metres
Specific Gravity of Water	1.0250
Mean Shell Thickness	0.0100 metres
Longitudinal Datum	FR NO. 15 (27MTRS FROM FR NO. 0)
Vertical Datum	BASELINE
Trim Length	36.000 metres

Draught Marks	Name	X metres	Z metres
Aft Marks	FR NO. 05	-18.000	-0.010
Mid Marks	Midships	0.000	-0.010
Fwd Marks	FR NO. 25	18.000	-0.010

Trim Between Marks 0.000 metres

Draught at Mid Marks metres	Moulded Draught metres	Moulded Displacement tonnes	Full Displacement tonnes	LCB metres	LCF metres	Moulded VCB metres	Immersion tonnes/cm	WSA metres²
0.500	0.490	342.012	349.845	0.946	0.836	0.252	7.590	764.22
0.550	0.540	380.084	387.990	0.935	0.836	0.279	7.639	771.37
0.600	0.590	418.397	426.377	0.926	0.836	0.305	7.687	778.53
0.650	0.640	456.953	465.006	0.918	0.837	0.331	7.735	785.68
0.700	0.690	495.680	503.792	0.912	0.837	0.357	7.748	791.41
0.750	0.740	534.433	542.600	0.906	0.837	0.383	7.753	796.77
0.800	0.790	573.206	581.428	0.901	0.837	0.409	7.757	802.14
0.850	0.840	612.000	620.276	0.897	0.837	0.435	7.761	807.50
0.900	0.890	650.814	659.146	0.893	0.837	0.460	7.765	812.87
0.950	0.940	689.649	698.036	0.890	0.835	0.486	7.768	818.21
1.000	0.990	729.023	737.666	0.863	0.275	0.512	7.952	843.27
1.050	1.040	769.298	778.198	0.855	0.894	0.538	8.158	868.33
1.100	1.090	810.093	819.048	0.857	0.895	0.565	8.160	873.66
1.150	1.140	850.902	859.912	0.858	0.895	0.591	8.163	878.99
1.200	1.190	891.725	900.789	0.860	0.895	0.617	8.166	884.32
1.250	1.240	932.561	941.680	0.862	0.895	0.644	8.169	889.65
1.300	1.290	973.410	982.584	0.863	0.895	0.670	8.171	894.98
1.350	1.340	1014.273	1023.501	0.864	0.896	0.696	8.174	900.30
1.400	1.390	1055.149	1064.432	0.865	0.896	0.722	8.177	905.63
1.450	1.440	1096.039	1105.376	0.866	0.896	0.747	8.179	910.96
1.500	1.490	1136.942	1146.334	0.867	0.896	0.773	8.182	916.29
1.550	1.540	1177.859	1187.305	0.868	0.897	0.799	8.185	921.62
1.600	1.590	1219.146	1229.014	0.854	-0.194	0.825	8.545	962.74
1.650	1.640	1261.881	1271.806	0.818	-0.199	0.852	8.549	968.27
1.700	1.690	1304.635	1314.616	0.785	-0.204	0.878	8.552	973.76
1.750	1.740	1347.402	1357.439	0.754	-0.211	0.905	8.555	979.21
1.800	1.790	1390.580	1400.980	0.737	0.724	0.932	8.864	1014.62
1.850	1.840	1434.911	1445.370	0.737	0.723	0.959	8.868	1020.36
1.900	1.890	1479.265	1489.782	0.736	0.721	0.986	8.873	1026.10
1.950	1.940	1523.641	1534.218	0.736	0.720	1.013	8.878	1031.84
2.000	1.990	1568.041	1578.676	0.736	0.719	1.040	8.882	1037.59
2.050	2.040	1612.433	1623.132	0.735	0.737	1.067	8.880	1043.81

2.100	2.090	1656.847	1667.603	0.735	0.736	1.094	8.885	1049.41
2.150	2.140	1701.283	1712.097	0.735	0.735	1.120	8.890	1055.02
2.200	2.190	1745.743	1756.614	0.735	0.733	1.147	8.894	1060.59
2.250	2.240	1790.645	1801.679	0.729	0.496	1.174	9.004	1076.46
2.300	2.290	1835.670	1846.761	0.724	0.501	1.201	9.006	1082.07
2.350	2.340	1880.708	1891.857	0.719	0.506	1.227	9.009	1087.68
2.400	2.390	1925.761	1936.967	0.714	0.510	1.254	9.012	1093.29
2.450	2.440	1970.827	1982.091	0.709	0.513	1.280	9.014	1098.89
2.500	2.490	2015.900	2027.220	0.705	0.511	1.307	9.015	1104.41
2.550	2.540	2060.976	2072.353	0.701	0.510	1.333	9.016	1109.94
2.600	2.590	2106.056	2117.489	0.696	0.508	1.360	9.016	1115.47
2.650	2.640	2151.139	2162.629	0.692	0.507	1.386	9.017	1121.00
2.700	2.690	2196.226	2207.773	0.689	0.505	1.412	9.018	1126.53
2.750	2.740	2241.316	2252.920	0.685	0.503	1.438	9.018	1132.06
2.800	2.790	2286.410	2298.071	0.681	0.502	1.465	9.019	1137.59
2.850	2.840	2331.508	2343.225	0.678	0.500	1.491	9.020	1143.12
2.900	2.890	2376.608	2388.381	0.675	0.500	1.517	9.020	1148.62
2.950	2.940	2421.708	2433.538	0.671	0.500	1.543	9.020	1154.12
3.000	2.990	2466.808	2478.694	0.668	0.500	1.569	9.020	1159.62

Trim Between Marks 0.000 metres

Draught at Mid Marks metres	Moulded KMT metres	Moulded KML metres	MCT tonnes-metres/cm	CB	CP	CM	CW
0.500	45.304	412.648	39.203	0.774	0.815	0.949	0.841
0.550	41.601	373.731	39.458	0.780	0.819	0.952	0.847
0.600	38.564	341.708	39.714	0.786	0.823	0.956	0.852
0.650	36.030	314.895	39.970	0.792	0.826	0.959	0.858
0.700	33.431	291.046	40.074	0.796	0.828	0.962	0.859
0.750	31.105	270.387	40.140	0.801	0.830	0.964	0.859
0.800	29.097	252.516	40.207	0.804	0.832	0.967	0.860
0.850	27.346	236.904	40.274	0.808	0.834	0.969	0.860
0.900	25.807	223.150	40.341	0.811	0.835	0.970	0.861
0.950	24.437	210.891	40.400	0.813	0.837	0.972	0.861
1.000	23.668	214.009	43.338	0.816	0.839	0.973	0.882
1.050	22.972	219.020	46.803	0.820	0.841	0.975	0.904
1.100	21.887	208.241	46.860	0.824	0.844	0.976	0.905
1.150	20.909	198.493	46.916	0.828	0.847	0.977	0.905
1.200	20.022	189.638	46.974	0.831	0.850	0.978	0.905
1.250	19.215	181.558	47.032	0.834	0.852	0.979	0.906
1.300	18.478	174.155	47.090	0.837	0.854	0.980	0.906
1.350	17.802	167.348	47.149	0.839	0.856	0.980	0.906
1.400	17.180	161.068	47.209	0.842	0.858	0.981	0.906
1.450	16.607	155.257	47.269	0.844	0.860	0.982	0.907
1.500	16.077	149.864	47.330	0.846	0.861	0.982	0.907
1.550	15.585	144.845	47.391	0.848	0.863	0.983	0.907
1.600	15.660	159.249	53.930	0.850	0.864	0.983	0.947
1.650	15.205	154.127	54.025	0.853	0.867	0.984	0.948
1.700	14.775	149.282	54.100	0.856	0.869	0.984	0.948
1.750	14.371	144.713	54.163	0.859	0.872	0.985	0.948
1.800	14.388	156.164	60.322	0.861	0.874	0.985	0.983
1.850	14.017	151.611	60.430	0.865	0.877	0.986	0.983
1.900	13.671	147.330	60.539	0.868	0.880	0.986	0.984
1.950	13.345	143.298	60.648	0.871	0.883	0.986	0.984
2.000	13.040	139.493	60.758	0.874	0.885	0.987	0.985

2.050	12.753	135.616	60.742	0.876	0.888	0.987	0.985
2.100	12.483	132.223	60.854	0.879	0.890	0.987	0.985
2.150	12.229	129.006	60.966	0.881	0.892	0.988	0.986
2.200	11.988	125.970	61.086	0.884	0.895	0.988	0.986
2.250	11.864	127.484	63.411	0.886	0.897	0.988	0.998
2.300	11.638	124.521	63.494	0.889	0.899	0.988	0.998
2.350	11.424	121.701	63.579	0.891	0.901	0.989	0.999
2.400	11.220	119.013	63.664	0.893	0.903	0.989	0.999
2.450	11.026	116.429	63.739	0.895	0.905	0.989	0.999
2.500	10.837	113.908	63.785	0.898	0.907	0.989	0.999
2.550	10.657	111.498	63.832	0.900	0.909	0.990	1.000
2.600	10.486	109.191	63.879	0.901	0.911	0.990	1.000
2.650	10.323	106.983	63.926	0.903	0.912	0.990	1.000
2.700	10.167	104.866	63.975	0.905	0.914	0.990	1.000
2.750	10.019	102.835	64.024	0.907	0.916	0.990	1.000
2.800	9.878	100.885	64.073	0.909	0.917	0.991	1.000
2.850	9.744	99.011	64.123	0.910	0.919	0.991	1.000
2.900	9.614	97.191	64.162	0.912	0.920	0.991	1.000
2.950	9.489	95.435	64.199	0.913	0.921	0.991	1.000
3.000	9.369	93.744	64.236	0.915	0.923	0.991	1.000

Trim Between Marks 0.550 metres by the bow

Draught at Mid Marks metres	Moulded Draught metres	Moulded Displacement tonnes	Full Displacement tonnes	LCB metres	LCF metres	Moulded VCB metres	Immersion tonnes/cm	WSA metres²
0.500	0.490	359.076	366.583	6.523	2.066	0.306	7.286	732.38
0.550	0.540	395.588	403.160	6.117	2.039	0.328	7.318	738.74
0.600	0.590	432.251	439.887	5.775	2.007	0.351	7.346	744.93
0.650	0.640	469.247	477.147	5.498	2.595	0.374	7.573	770.80
0.700	0.690	507.744	516.016	5.237	1.636	0.399	7.898	806.98
0.750	0.740	547.287	555.622	4.978	1.604	0.424	7.918	813.14
0.800	0.790	586.923	595.320	4.752	1.576	0.448	7.936	819.16
0.850	0.840	626.636	635.092	4.552	1.552	0.473	7.949	825.04
0.900	0.890	666.408	674.923	4.374	1.534	0.498	7.959	830.79
0.950	0.940	706.223	714.796	4.215	1.523	0.523	7.966	836.39
1.000	0.990	746.064	754.693	4.072	1.520	0.548	7.970	841.89
1.050	1.040	785.923	794.609	3.944	1.518	0.573	7.974	847.36
1.100	1.090	825.800	834.541	3.828	1.516	0.597	7.977	852.83
1.150	1.140	865.693	874.491	3.722	1.514	0.622	7.980	858.30
1.200	1.190	905.604	914.458	3.625	1.512	0.647	7.984	863.78
1.250	1.240	945.532	954.442	3.537	1.510	0.672	7.987	869.25
1.300	1.290	985.478	994.444	3.455	1.508	0.697	7.991	874.71
1.350	1.340	1025.825	1035.049	3.364	0.950	0.722	8.174	899.88
1.400	1.390	1067.478	1077.064	3.296	1.890	0.748	8.482	935.18
1.450	1.440	1109.898	1119.540	3.243	1.893	0.775	8.486	940.74
1.500	1.490	1152.334	1162.034	3.193	1.895	0.801	8.489	946.30
1.550	1.540	1194.787	1204.544	3.147	1.897	0.828	8.492	951.86
1.600	1.590	1237.258	1247.071	3.105	1.899	0.854	8.496	957.42
1.650	1.640	1279.745	1289.615	3.065	1.902	0.880	8.499	962.98
1.700	1.690	1322.249	1332.176	3.028	1.904	0.906	8.503	968.54
1.750	1.740	1364.770	1374.755	2.993	1.906	0.932	8.506	974.10
1.800	1.790	1407.333	1417.389	2.961	1.952	0.959	8.523	981.04
1.850	1.840	1449.959	1460.072	2.932	1.954	0.985	8.527	986.61

1.900	1.890	1492.602	1502.772	2.904	1.957	1.011	8.530	992.18
1.950	1.940	1535.263	1545.490	2.878	1.959	1.037	8.534	997.76
2.000	1.990	1578.834	1589.485	2.831	0.870	1.063	8.895	1039.17
2.050	2.040	1623.318	1634.028	2.777	0.866	1.089	8.898	1044.89
2.100	2.090	1667.816	1678.585	2.726	0.859	1.116	8.901	1050.56
2.150	2.140	1712.328	1723.154	2.678	0.852	1.142	8.904	1056.23
2.200	2.190	1756.852	1767.736	2.632	0.845	1.168	8.906	1061.90
2.250	2.240	1801.388	1812.331	2.588	0.838	1.194	8.909	1067.57
2.300	2.290	1845.938	1856.939	2.545	0.831	1.221	8.911	1073.24
2.350	2.340	1890.500	1901.559	2.505	0.825	1.247	8.914	1078.91
2.400	2.390	1935.075	1946.192	2.466	0.818	1.273	8.916	1084.58
2.450	2.440	1979.636	1990.816	2.430	0.830	1.299	8.913	1090.74
2.500	2.490	2024.205	2035.442	2.395	0.822	1.325	8.915	1096.27
2.550	2.540	2068.787	2080.080	2.361	0.815	1.351	8.918	1101.80
2.600	2.590	2113.270	2124.969	2.331	-0.260	1.376	8.572	1141.30
2.650	2.640	2155.829	2167.893	2.271	-1.219	1.401	8.442	1176.96
2.700	2.690	2195.802	2208.580	2.190	-3.246	1.423	7.779	1246.66
2.750	2.740	2233.218	2246.646	2.089	-5.085	1.443	7.177	1310.05
2.800	2.790	2267.972	2282.034	1.973	-6.887	1.462	6.587	1371.92
2.850	2.840	2300.072	2314.765	1.845	-8.689	1.480	5.997	1433.46
2.900	2.890	2329.519	2344.839	1.710	-10.491	1.495	5.407	1494.68
2.950	2.940	2356.114	2371.665	1.570	-11.094	1.510	5.211	1517.15
3.000	2.990	2380.069	2396.388	1.431	-13.341	1.522	4.474	1592.10

Trim Between Marks 0.550 metres by the bow

Draught at Mid Marks metres	Moulded KMT metres	Moulded KML metres	MCT tonnes-metres/cm	CB	CP	CM	CW
0.500	41.710	347.471	34.658	0.812	0.856	0.949	0.808
0.550	38.409	316.736	34.805	0.812	0.853	0.952	0.811
0.600	35.595	290.967	34.936	0.812	0.850	0.956	0.814
0.650	33.930	291.706	38.023	0.813	0.848	0.959	0.840
0.700	32.760	304.151	42.897	0.816	0.848	0.962	0.876
0.750	30.673	283.254	43.061	0.820	0.850	0.964	0.878
0.800	28.832	265.101	43.221	0.824	0.852	0.967	0.880
0.850	27.191	249.154	43.369	0.827	0.854	0.969	0.881
0.900	25.714	234.985	43.499	0.830	0.856	0.970	0.882
0.950	24.376	222.256	43.601	0.833	0.857	0.972	0.883
1.000	23.159	210.724	43.670	0.835	0.858	0.973	0.884
1.050	22.063	200.330	43.734	0.838	0.860	0.975	0.884
1.100	21.075	190.937	43.799	0.840	0.861	0.976	0.884
1.150	20.181	182.408	43.864	0.842	0.862	0.977	0.885
1.200	19.368	174.630	43.929	0.844	0.863	0.978	0.885
1.250	18.625	167.507	43.995	0.845	0.864	0.979	0.886
1.300	17.943	160.945	44.058	0.847	0.865	0.980	0.886
1.350	17.638	165.475	47.152	0.849	0.866	0.980	0.906
1.400	17.532	177.979	52.775	0.851	0.868	0.981	0.940
1.450	16.934	171.426	52.851	0.855	0.870	0.982	0.941
1.500	16.382	165.354	52.929	0.857	0.873	0.982	0.941
1.550	15.871	159.714	53.007	0.860	0.875	0.983	0.942
1.600	15.396	154.460	53.085	0.863	0.877	0.983	0.942
1.650	14.955	149.554	53.164	0.865	0.879	0.984	0.942
1.700	14.544	144.964	53.244	0.867	0.881	0.984	0.943
1.750	14.160	140.659	53.324	0.870	0.883	0.985	0.943
1.800	13.819	137.293	53.671	0.872	0.885	0.985	0.945

1.850	13.481	133.464	53.755	0.874	0.886	0.986	0.945
1.900	13.165	129.853	53.839	0.876	0.888	0.986	0.946
1.950	12.867	126.443	53.923	0.877	0.889	0.986	0.946
2.000	12.995	139.185	61.042	0.880	0.891	0.987	0.986
2.050	12.707	135.577	61.134	0.882	0.894	0.987	0.987
2.100	12.432	132.119	61.208	0.885	0.896	0.987	0.987
2.150	12.173	128.842	61.283	0.887	0.898	0.988	0.987
2.200	11.928	125.731	61.358	0.889	0.900	0.988	0.987
2.250	11.696	122.774	61.434	0.892	0.902	0.988	0.988
2.300	11.477	119.960	61.511	0.894	0.904	0.988	0.988
2.350	11.269	117.279	61.588	0.896	0.906	0.989	0.988
2.400	11.073	114.722	61.665	0.898	0.908	0.989	0.989
2.450	10.886	112.048	61.615	0.899	0.909	0.989	0.988
2.500	10.708	109.718	61.692	0.901	0.911	0.989	0.988
2.550	10.539	107.489	61.770	0.903	0.912	0.990	0.989
2.600	10.027	93.553	54.917	0.905	0.914	0.990	0.950
2.650	9.741	87.790	52.573	0.905	0.914	0.990	0.936
2.700	8.968	67.761	41.331	0.905	0.914	0.990	0.862
2.750	8.288	52.671	32.674	0.904	0.912	0.990	0.796
2.800	7.648	40.455	25.486	0.901	0.910	0.991	0.730
2.850	7.033	30.490	19.480	0.898	0.906	0.991	0.665
2.900	6.440	22.484	14.549	0.894	0.902	0.991	0.599
2.950	6.221	20.079	13.141	0.888	0.897	0.991	0.578
3.000	5.527	13.171	8.708	0.882	0.890	0.991	0.496

Trim Between Marks 0.550 metres by the stern

Draught at Mid Marks metres	Moulded Draught metres	Moulded Displacement tonnes	Full Displacement tonnes	LCB metres	LCF metres	Moulded VCB metres	Immersion tonnes/cm	WSA metres²
0.500	0.490	338.619	346.166	-5.016	-0.432	0.295	7.261	736.31
0.550	0.540	375.012	382.624	-4.578	-0.406	0.317	7.295	742.69
0.600	0.590	411.567	419.244	-4.212	-0.377	0.340	7.326	748.93
0.650	0.640	449.191	457.438	-3.905	0.043	0.363	7.841	804.62
0.700	0.690	488.465	496.775	-3.591	0.075	0.388	7.867	810.77
0.750	0.740	527.859	536.231	-3.320	0.109	0.412	7.890	816.75
0.800	0.790	567.356	575.787	-3.083	0.140	0.436	7.908	822.59
0.850	0.840	606.937	615.428	-2.875	0.167	0.461	7.924	828.30
0.900	0.890	646.588	655.135	-2.689	0.189	0.486	7.936	833.87
0.950	0.940	686.289	694.892	-2.524	0.204	0.510	7.944	839.30
1.000	0.990	726.026	734.683	-2.376	0.212	0.535	7.950	844.60
1.050	1.040	765.783	774.494	-2.243	0.214	0.560	7.953	849.83
1.100	1.090	805.557	814.322	-2.123	0.216	0.584	7.957	855.05
1.150	1.140	845.348	854.166	-2.014	0.218	0.609	7.960	860.27
1.200	1.190	885.156	894.028	-1.914	0.221	0.634	7.963	865.49
1.250	1.240	926.592	935.885	-1.873	-0.874	0.661	8.326	906.67
1.300	1.290	968.233	977.582	-1.830	-0.877	0.687	8.330	912.08
1.350	1.340	1009.890	1019.294	-1.791	-0.884	0.714	8.333	917.39
1.400	1.390	1051.560	1061.018	-1.756	-0.891	0.740	8.335	922.71
1.450	1.440	1094.057	1103.774	-1.699	-0.274	0.766	8.542	948.02
1.500	1.490	1136.777	1146.550	-1.645	-0.277	0.793	8.546	953.53
1.550	1.540	1179.515	1189.345	-1.596	-0.280	0.819	8.550	959.04
1.600	1.590	1222.273	1232.166	-1.551	-0.264	0.845	8.547	965.15
1.650	1.640	1265.019	1274.967	-1.508	-0.267	0.872	8.551	970.53

1.700	1.690	1307.785	1317.788	-1.467	-0.270	0.898	8.555	975.90
1.750	1.740	1350.569	1360.627	-1.430	-0.273	0.924	8.559	981.28
1.800	1.790	1393.396	1403.600	-1.397	-0.557	0.950	8.653	995.44
1.850	1.840	1436.675	1446.934	-1.371	-0.561	0.976	8.658	1000.91
1.900	1.890	1479.971	1490.285	-1.348	-0.560	1.002	8.660	1006.29
1.950	1.940	1523.277	1533.646	-1.326	-0.558	1.028	8.662	1011.66
2.000	1.990	1566.593	1577.017	-1.304	-0.555	1.055	8.664	1017.03
2.050	2.040	1609.919	1620.398	-1.284	-0.553	1.081	8.666	1022.39
2.100	2.090	1653.252	1663.787	-1.265	-0.554	1.107	8.667	1027.72
2.150	2.140	1696.589	1707.178	-1.247	-0.556	1.133	8.668	1033.03
2.200	2.190	1739.945	1750.894	-1.225	0.383	1.159	8.975	1068.22
2.250	2.240	1784.824	1795.831	-1.185	0.387	1.185	8.977	1073.82
2.300	2.290	1829.717	1840.782	-1.146	0.391	1.211	8.980	1079.42
2.350	2.340	1874.624	1885.746	-1.109	0.395	1.238	8.983	1085.02
2.400	2.390	1919.544	1930.723	-1.074	0.400	1.264	8.985	1090.62
2.450	2.440	1964.478	1975.714	-1.041	0.404	1.290	8.988	1096.22
2.500	2.490	2009.423	2020.717	-1.008	0.410	1.316	8.990	1101.80
2.550	2.540	2054.379	2065.730	-0.978	0.416	1.342	8.992	1107.37
2.600	2.590	2099.321	2110.827	-0.949	0.726	1.368	8.896	1122.54
2.650	2.640	2142.839	2154.808	-0.907	1.975	1.393	8.518	1167.71
2.700	2.690	2183.779	2196.342	-0.840	3.628	1.416	7.979	1225.60
2.750	2.740	2221.957	2235.109	-0.750	5.289	1.437	7.440	1283.15
2.800	2.790	2257.492	2271.280	-0.643	7.095	1.457	6.852	1345.17
2.850	2.840	2290.380	2304.801	-0.524	8.902	1.474	6.264	1406.87
2.900	2.890	2320.618	2335.667	-0.395	10.703	1.491	5.674	1468.16
2.950	2.940	2348.101	2363.578	-0.261	11.904	1.505	5.280	1509.92
3.000	2.990	2372.928	2389.026	-0.125	13.705	1.519	4.690	1570.55

Trim Between Marks 0.550 metres by the stern

Draught at Mid Marks metres	Moulded KMT metres	Moulded KML metres	MCT tonnes-metres/cm	LCG metres	CP	CM	CW
0.500	43.805	365.931	34.420	-5.020	0.807	0.949	0.805
0.550	40.156	331.873	34.571	-4.583	0.808	0.952	0.809
0.600	37.094	303.682	34.718	-4.217	0.809	0.956	0.812
0.650	36.423	339.044	42.304	-3.911	0.812	0.959	0.869
0.700	33.874	313.090	42.482	-3.597	0.816	0.962	0.872
0.750	31.654	290.850	42.647	-3.326	0.820	0.964	0.875
0.800	29.706	271.630	42.809	-3.090	0.824	0.967	0.877
0.850	27.977	254.831	42.963	-2.882	0.827	0.969	0.878
0.900	26.428	239.985	43.103	-2.697	0.830	0.970	0.880
0.950	25.029	226.722	43.221	-2.532	0.833	0.972	0.881
1.000	23.757	214.742	43.308	-2.384	0.835	0.973	0.881
1.050	22.603	203.886	43.370	-2.252	0.838	0.975	0.882
1.100	21.565	194.101	43.433	-2.132	0.840	0.976	0.882
1.150	20.627	185.235	43.497	-2.023	0.842	0.977	0.882
1.200	19.775	177.165	43.561	-1.924	0.843	0.978	0.883
1.250	19.684	193.460	49.794	-1.883	0.846	0.979	0.923
1.300	18.920	185.469	49.882	-1.841	0.850	0.980	0.924
1.350	18.208	178.017	49.938	-1.802	0.852	0.980	0.924
1.400	17.555	171.156	49.995	-1.767	0.855	0.981	0.924
1.450	17.281	177.115	53.826	-1.710	0.858	0.982	0.947
1.500	16.706	170.727	53.911	-1.658	0.861	0.982	0.947
1.550	16.175	164.802	53.996	-1.609	0.864	0.983	0.948
1.600	15.682	158.943	53.965	-1.564	0.867	0.983	0.948

1.650	15.225	153.819	54.051	-1.521	0.869	0.984	0.948
1.700	14.799	149.029	54.138	-1.481	0.872	0.984	0.948
1.750	14.402	144.542	54.226	-1.444	0.874	0.985	0.949
1.800	14.148	144.934	56.097	-1.411	0.876	0.985	0.959
1.850	13.796	140.820	56.198	-1.386	0.878	0.986	0.960
1.900	13.456	136.857	56.262	-1.363	0.880	0.986	0.960
1.950	13.136	133.109	56.323	-1.341	0.883	0.986	0.960
2.000	12.835	129.569	56.384	-1.321	0.885	0.987	0.961
2.050	12.552	126.220	56.446	-1.301	0.886	0.987	0.961
2.100	12.280	123.003	56.488	-1.282	0.888	0.987	0.961
2.150	12.023	119.941	56.525	-1.265	0.890	0.988	0.961
2.200	12.097	129.943	62.804	-1.243	0.892	0.988	0.995
2.250	11.856	126.840	62.885	-1.203	0.894	0.988	0.995
2.300	11.629	123.889	62.967	-1.165	0.896	0.988	0.996
2.350	11.414	121.080	63.050	-1.128	0.898	0.989	0.996
2.400	11.211	118.403	63.133	-1.094	0.900	0.989	0.996
2.450	11.017	115.847	63.216	-1.060	0.902	0.989	0.996
2.500	10.832	113.384	63.288	-1.029	0.904	0.989	0.997
2.550	10.656	111.029	63.360	-0.998	0.906	0.990	0.997
2.600	10.388	105.214	61.355	-0.970	0.908	0.990	0.986
2.650	9.856	90.719	53.999	-0.928	0.909	0.990	0.944
2.700	9.198	73.514	44.594	-0.861	0.909	0.990	0.885
2.750	8.571	58.821	36.305	-0.772	0.908	0.990	0.825
2.800	7.927	45.561	28.571	-0.665	0.906	0.991	0.760
2.850	7.307	34.673	22.059	-0.546	0.902	0.991	0.694
2.900	6.705	25.830	16.651	-0.418	0.898	0.991	0.629
2.950	6.301	20.891	13.626	-0.284	0.893	0.991	0.585
3.000	5.734	14.953	9.856	-0.148	0.888	0.991	0.520

Free Trimming Cross Curve Data

55M Pontoon

Specific Gravity of Water 1.0250
Mean Shell Thickness 0.0100 metres
Longitudinal Datum FR NO. 15 (27MTRS FROM FR NO. 0)
Vertical Datum BASELINE
Trim Length 36.000 metres

Draught Marks	**Name**	**X metres**	**Z metres**
Aft Marks	FR NO. 05	-18.000	-0.010
Mid Marks	Midships	0.000	-0.010
Fwd Marks	FR NO. 25	18.000	-0.010

Trim Between Marks 0.000 metres

Displacement				**Heel Angle - Degrees / KN - metres**						
tonnes	**0.00**	**10.00**	**20.00**	**30.00**	**40.00**	**50.00**	**60.00**	**70.00**	**80.00**	**90.00**
340.000	0.000	4.946	5.810	6.005	5.832	5.336	4.616	3.726	2.709	1.604
390.000	0.000	4.760	5.708	5.950	5.764	5.268	4.557	3.683	2.684	1.599
440.000	0.000	4.584	5.611	5.883	5.684	5.191	4.493	3.637	2.658	1.595
490.000	0.000	4.421	5.520	5.802	5.595	5.110	4.427	3.589	2.632	1.592
540.000	0.000	4.267	5.434	5.709	5.500	5.025	4.358	3.540	2.606	1.589
590.000	0.000	4.121	5.350	5.610	5.400	4.936	4.287	3.490	2.579	1.587
640.000	0.000	3.982	5.266	5.504	5.297	4.846	4.215	3.440	2.553	1.585
690.000	0.000	3.848	5.174	5.394	5.190	4.754	4.141	3.389	2.526	1.584
740.000	0.000	3.720	5.074	5.281	5.083	4.660	4.067	3.337	2.499	1.582
790.000	0.000	3.596	4.969	5.164	4.973	4.565	3.992	3.285	2.472	1.581
840.000	0.000	3.477	4.858	5.044	4.862	4.470	3.916	3.233	2.445	1.580
890.000	0.000	3.361	4.742	4.922	4.750	4.373	3.841	3.180	2.418	1.579
940.000	0.000	3.248	4.623	4.798	4.637	4.277	3.764	3.127	2.390	1.578
990.000	0.000	3.138	4.502	4.673	4.522	4.179	3.688	3.075	2.363	1.578
1040.000	0.000	3.031	4.378	4.547	4.406	4.081	3.611	3.022	2.336	1.577
1090.000	0.000	2.931	4.252	4.420	4.290	3.983	3.534	2.969	2.308	1.577
1140.000	0.000	2.839	4.124	4.292	4.174	3.883	3.456	2.915	2.281	1.576
1190.000	0.000	2.754	3.994	4.163	4.058	3.784	3.378	2.862	2.254	1.575
1240.000	0.000	2.674	3.862	4.035	3.942	3.686	3.300	2.807	2.226	1.575
1290.000	0.000	2.599	3.730	3.905	3.824	3.587	3.223	2.754	2.197	1.573
1340.000	0.000	2.522	3.596	3.776	3.707	3.488	3.146	2.701	2.170	1.572
1390.000	0.000	2.444	3.461	3.645	3.590	3.388	3.068	2.648	2.143	1.572
1440.000	0.000	2.364	3.325	3.514	3.472	3.288	2.990	2.594	2.116	1.572
1490.000	0.000	2.282	3.189	3.382	3.354	3.189	2.912	2.541	2.089	1.572
1540.000	0.000	2.198	3.051	3.250	3.236	3.089	2.834	2.487	2.061	1.571
1590.000	0.000	2.113	2.912	3.118	3.117	2.989	2.756	2.433	2.034	1.571
1640.000	0.000	2.026	2.773	2.985	2.998	2.888	2.678	2.380	2.006	1.571
1690.000	0.000	1.938	2.633	2.852	2.879	2.788	2.599	2.326	1.979	1.571
1740.000	0.000	1.849	2.493	2.719	2.760	2.688	2.521	2.272	1.952	1.571
1790.000	0.000	1.758	2.354	2.586	2.641	2.587	2.443	2.218	1.924	1.570
1840.000	0.000	1.667	2.217	2.452	2.522	2.487	2.364	2.165	1.897	1.570
1890.000	0.000	1.574	2.083	2.318	2.402	2.386	2.286	2.111	1.869	1.570
1940.000	0.000	1.480	1.950	2.184	2.283	2.285	2.207	2.057	1.842	1.570
1990.000	0.000	1.385	1.819	2.050	2.163	2.184	2.128	2.003	1.815	1.570
2040.000	0.000	1.288	1.688	1.917	2.043	2.083	2.050	1.949	1.787	1.570
2090.000	0.000	1.189	1.559	1.786	1.923	1.982	1.971	1.895	1.760	1.569
2140.000	0.000	1.088	1.432	1.657	1.803	1.881	1.892	1.841	1.732	1.569
2190.000	0.000	0.985	1.306	1.530	1.686	1.780	1.814	1.787	1.705	1.569
2240.000	0.000	0.880	1.181	1.404	1.570	1.680	1.735	1.733	1.677	1.569

2290.000	0.000	0.771	1.057	1.280	1.455	1.581	1.656	1.679	1.650	1.569
2340.000	0.000	0.659	0.933	1.158	1.343	1.485	1.580	1.626	1.622	1.569
2390.000	0.000	0.543	0.809	1.039	1.234	1.390	1.505	1.574	1.595	1.569
2440.000	0.000	0.421	0.684	0.920	1.128	1.300	1.434	1.525	1.571	1.570

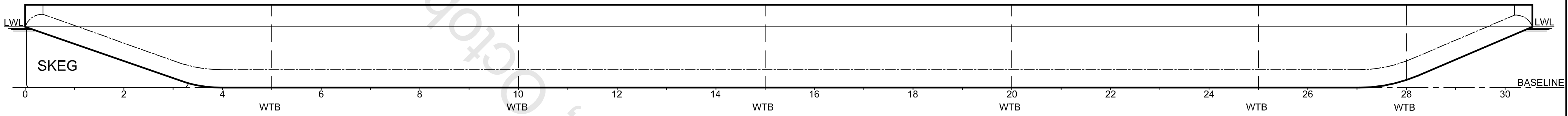
Trim Between Marks 0.550 metres by the bow

Displacement	Heel Angle - Degrees / KN - metres									
tonnes	**0.00**	**10.00**	**20.00**	**30.00**	**40.00**	**50.00**	**60.00**	**70.00**	**80.00**	**90.00**
340.000	0.000	4.790	5.737	5.925	5.709	5.218	4.520	3.659	2.673	1.599
390.000	0.000	4.632	5.642	5.847	5.638	5.155	4.468	3.621	2.651	1.595
440.000	0.000	4.478	5.548	5.765	5.560	5.085	4.410	3.579	2.627	1.591
490.000	0.000	4.329	5.453	5.676	5.474	5.008	4.347	3.534	2.603	1.588
540.000	0.000	4.186	5.360	5.585	5.385	4.929	4.283	3.488	2.578	1.586
590.000	0.000	4.047	5.265	5.489	5.291	4.845	4.216	3.441	2.553	1.584
640.000	0.000	3.913	5.169	5.388	5.194	4.759	4.147	3.393	2.528	1.583
690.000	0.000	3.783	5.070	5.283	5.092	4.671	4.076	3.344	2.502	1.581
740.000	0.000	3.658	4.967	5.173	4.988	4.581	4.005	3.294	2.476	1.580
790.000	0.000	3.537	4.862	5.060	4.882	4.489	3.932	3.244	2.450	1.579
840.000	0.000	3.420	4.752	4.944	4.774	4.396	3.859	3.193	2.424	1.578
890.000	0.000	3.308	4.639	4.826	4.664	4.302	3.785	3.142	2.397	1.577
940.000	0.000	3.200	4.522	4.705	4.553	4.207	3.710	3.090	2.370	1.577
990.000	0.000	3.097	4.402	4.582	4.441	4.112	3.635	3.038	2.344	1.576
1040.000	0.000	2.998	4.280	4.459	4.329	4.016	3.560	2.987	2.317	1.575
1090.000	0.000	2.902	4.154	4.333	4.214	3.919	3.484	2.934	2.290	1.575
1140.000	0.000	2.809	4.025	4.206	4.099	3.820	3.407	2.881	2.263	1.574
1190.000	0.000	2.719	3.895	4.076	3.982	3.722	3.330	2.828	2.236	1.574
1240.000	0.000	2.631	3.763	3.946	3.865	3.623	3.252	2.775	2.209	1.573
1290.000	0.000	2.544	3.630	3.816	3.748	3.523	3.175	2.721	2.181	1.573
1340.000	0.000	2.458	3.495	3.685	3.630	3.424	3.097	2.668	2.154	1.573
1390.000	0.000	2.373	3.360	3.554	3.511	3.324	3.019	2.614	2.127	1.572
1440.000	0.000	2.288	3.224	3.422	3.393	3.223	2.941	2.561	2.099	1.572
1490.000	0.000	2.203	3.086	3.290	3.274	3.123	2.862	2.507	2.072	1.572
1540.000	0.000	2.117	2.949	3.157	3.155	3.022	2.783	2.452	2.044	1.572
1590.000	0.000	2.032	2.813	3.027	3.038	2.923	2.705	2.399	2.016	1.571
1640.000	0.000	1.947	2.679	2.896	2.921	2.824	2.628	2.346	1.989	1.570
1690.000	0.000	1.861	2.545	2.766	2.804	2.726	2.551	2.293	1.962	1.569
1740.000	0.000	1.773	2.412	2.635	2.687	2.627	2.474	2.240	1.935	1.569
1790.000	0.000	1.684	2.279	2.504	2.570	2.528	2.397	2.187	1.908	1.569
1840.000	0.000	1.593	2.147	2.374	2.452	2.429	2.320	2.134	1.881	1.569
1890.000	0.000	1.500	2.015	2.244	2.335	2.330	2.242	2.081	1.854	1.569
1940.000	0.000	1.406	1.885	2.114	2.218	2.230	2.165	2.028	1.827	1.569
1990.000	0.000	1.311	1.756	1.985	2.101	2.131	2.087	1.975	1.800	1.569
2040.000	0.000	1.213	1.627	1.857	1.984	2.033	2.010	1.922	1.773	1.568
2090.000	0.000	1.113	1.500	1.730	1.869	1.934	1.933	1.869	1.746	1.568
2140.000	0.000	1.011	1.374	1.605	1.755	1.837	1.857	1.817	1.719	1.568
2190.000	0.000	0.910	1.251	1.483	1.644	1.743	1.783	1.766	1.693	1.568
2240.000	0.000	0.810	1.132	1.364	1.536	1.651	1.711	1.716	1.668	1.568
2290.000	0.000	0.711	1.015	1.249	1.431	1.561	1.641	1.668	1.643	1.568
2340.000	0.000	0.612	0.901	1.136	1.327	1.473	1.571	1.620	1.619	1.568
2390.000	0.000	0.511	0.788	1.025	1.226	1.386	1.503	1.573	1.596	1.569
2440.000	0.000	0.407	0.675	0.915	1.125	1.300	1.436	1.527	1.572	1.570

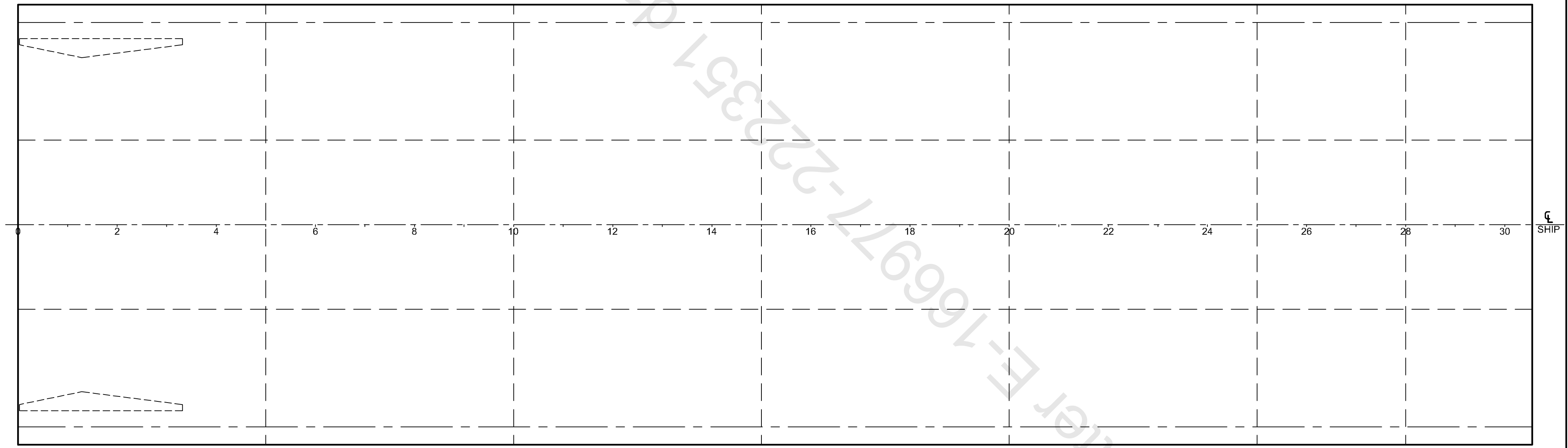
Trim Between Marks 0.550 metres by the stern

Displacement	Heel Angle - Degrees / KN - metres									
tonnes	**0.00**	**10.00**	**20.00**	**30.00**	**40.00**	**50.00**	**60.00**	**70.00**	**80.00**	**90.00**
340.000	0.000	4.796	5.753	5.944	5.728	5.237	4.538	3.674	2.685	1.608
390.000	0.000	4.640	5.656	5.867	5.657	5.174	4.485	3.635	2.662	1.602
440.000	0.000	4.487	5.560	5.783	5.579	5.103	4.426	3.592	2.638	1.598
490.000	0.000	4.342	5.466	5.698	5.496	5.028	4.364	3.548	2.613	1.595
540.000	0.000	4.202	5.372	5.607	5.407	4.949	4.300	3.502	2.588	1.592
590.000	0.000	4.066	5.279	5.511	5.314	4.865	4.233	3.454	2.563	1.590
640.000	0.000	3.935	5.183	5.409	5.216	4.779	4.163	3.406	2.537	1.588
690.000	0.000	3.807	5.086	5.303	5.114	4.690	4.093	3.356	2.511	1.586
740.000	0.000	3.682	4.985	5.193	5.009	4.600	4.021	3.306	2.485	1.585
790.000	0.000	3.562	4.881	5.080	4.901	4.507	3.948	3.256	2.458	1.583
840.000	0.000	3.444	4.774	4.965	4.793	4.413	3.874	3.205	2.432	1.582
890.000	0.000	3.332	4.663	4.847	4.683	4.318	3.799	3.153	2.405	1.581
940.000	0.000	3.222	4.544	4.724	4.570	4.221	3.722	3.099	2.377	1.580
990.000	0.000	3.117	4.423	4.599	4.456	4.124	3.645	3.045	2.348	1.578
1040.000	0.000	3.016	4.298	4.474	4.342	4.027	3.569	2.993	2.321	1.576
1090.000	0.000	2.920	4.172	4.348	4.227	3.929	3.492	2.940	2.294	1.576
1140.000	0.000	2.827	4.045	4.221	4.112	3.832	3.416	2.888	2.267	1.575
1190.000	0.000	2.736	3.916	4.094	3.997	3.734	3.340	2.835	2.240	1.575
1240.000	0.000	2.648	3.785	3.965	3.881	3.636	3.263	2.782	2.213	1.575
1290.000	0.000	2.561	3.653	3.836	3.764	3.537	3.185	2.729	2.186	1.574
1340.000	0.000	2.476	3.519	3.706	3.647	3.438	3.108	2.676	2.159	1.574
1390.000	0.000	2.391	3.384	3.576	3.530	3.339	3.031	2.623	2.131	1.574
1440.000	0.000	2.306	3.248	3.444	3.412	3.239	2.953	2.569	2.104	1.573
1490.000	0.000	2.220	3.110	3.312	3.293	3.139	2.874	2.515	2.077	1.573
1540.000	0.000	2.134	2.972	3.180	3.175	3.039	2.796	2.462	2.049	1.573
1590.000	0.000	2.047	2.834	3.047	3.056	2.938	2.718	2.408	2.022	1.572
1640.000	0.000	1.959	2.697	2.913	2.936	2.838	2.639	2.354	1.994	1.572
1690.000	0.000	1.870	2.559	2.780	2.817	2.737	2.561	2.300	1.967	1.572
1740.000	0.000	1.780	2.423	2.646	2.698	2.636	2.482	2.246	1.939	1.572
1790.000	0.000	1.689	2.289	2.515	2.580	2.537	2.404	2.193	1.912	1.572
1840.000	0.000	1.598	2.156	2.383	2.461	2.437	2.327	2.140	1.885	1.571
1890.000	0.000	1.505	2.023	2.252	2.343	2.337	2.249	2.087	1.858	1.571
1940.000	0.000	1.411	1.892	2.122	2.225	2.238	2.171	2.033	1.831	1.571
1990.000	0.000	1.315	1.762	1.993	2.108	2.138	2.093	1.980	1.803	1.571
2040.000	0.000	1.217	1.633	1.864	1.991	2.039	2.015	1.926	1.776	1.571
2090.000	0.000	1.116	1.505	1.736	1.875	1.940	1.938	1.873	1.749	1.571
2140.000	0.000	1.014	1.379	1.610	1.760	1.842	1.861	1.820	1.722	1.570
2190.000	0.000	0.912	1.255	1.487	1.648	1.747	1.787	1.769	1.696	1.570
2240.000	0.000	0.812	1.135	1.368	1.539	1.654	1.714	1.719	1.670	1.570
2290.000	0.000	0.712	1.018	1.252	1.433	1.564	1.643	1.670	1.645	1.570
2340.000	0.000	0.612	0.903	1.138	1.329	1.475	1.573	1.622	1.621	1.570
2390.000	0.000	0.511	0.789	1.026	1.227	1.387	1.504	1.574	1.597	1.570
2440.000	0.000	0.407	0.675	0.915	1.126	1.301	1.436	1.528	1.573	1.571

~~LOAD DEPARTURE CONDITION WITH CARGO~~ LIGHTSHIP



PROFILE



MAIN DECK PLAN

Condition 1: LIGHTSHIP CONDITION

Item	Weight	LCG	LMom	VCG	VMom	TCG	FSM
DECK CARGO	0.000	--	--	--	--	--	0.000
Deadweight	**0.000**	**0.000**	**0.00**	**0.000**	**0.00**	**0.000**	**0.000**
Lightship	**388.635**	**1.332**	**517.66**	**3.000**	**1165.90**	**0.000**	**0.000**
Displacement	**388.635**	**1.332**	**517.66**	**3.000**	**1165.90**	**0.000**	**0.000**

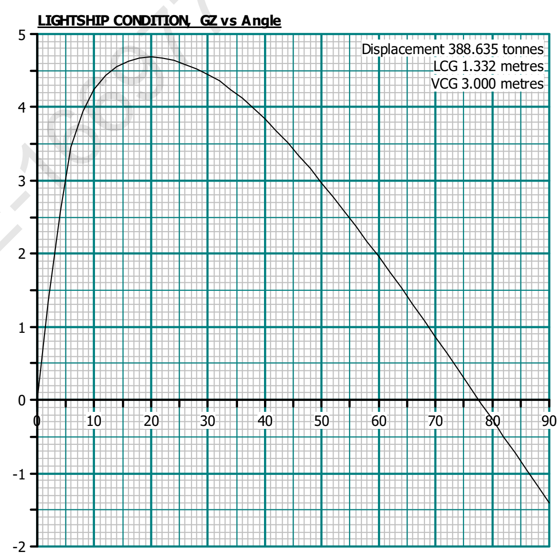
Draught		
	Aft	0.530 metres
	Mid	0.550 metres
	Fwd	0.570 metres

Trim Between Marks 0.039 metres by the bow

GM Solid 38.547 metres
GM Fluid 38.547 metres
Effective VCG 3.000 metres

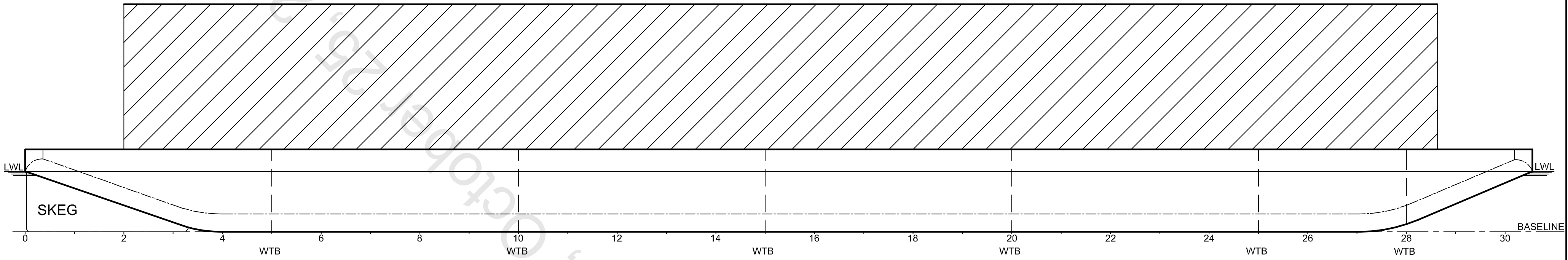
Moulded Displacement	380.730 tonnes
Waterline at LCF referred to hull definition datum	0.541 metres
LCF referred to hull definition datum	0.862 metres
Heel Angle	0.00 degrees to starboard

Heel Angle degrees	Righting GZ metres	Lever KN metres	Waterline metres	Trim metres	VCB metres	GZ Curve Area metres.rad
0.0	0.000	0.000	0.540	-0.039	0.279	0.000
10.0	4.243	4.764	0.306	-0.065	0.595	0.470
20.0	4.683	5.709	-0.394	-0.087	0.859	1.265
30.0	4.450	5.950	-1.243	-0.107	1.081	2.068
40.0	3.836	5.764	-2.080	-0.146	1.242	2.796
50.0	2.970	5.268	-2.823	-0.182	1.348	3.392
60.0	1.959	4.557	-3.450	-0.212	1.426	3.824
70.0	0.864	3.683	-3.942	-0.235	1.490	4.071

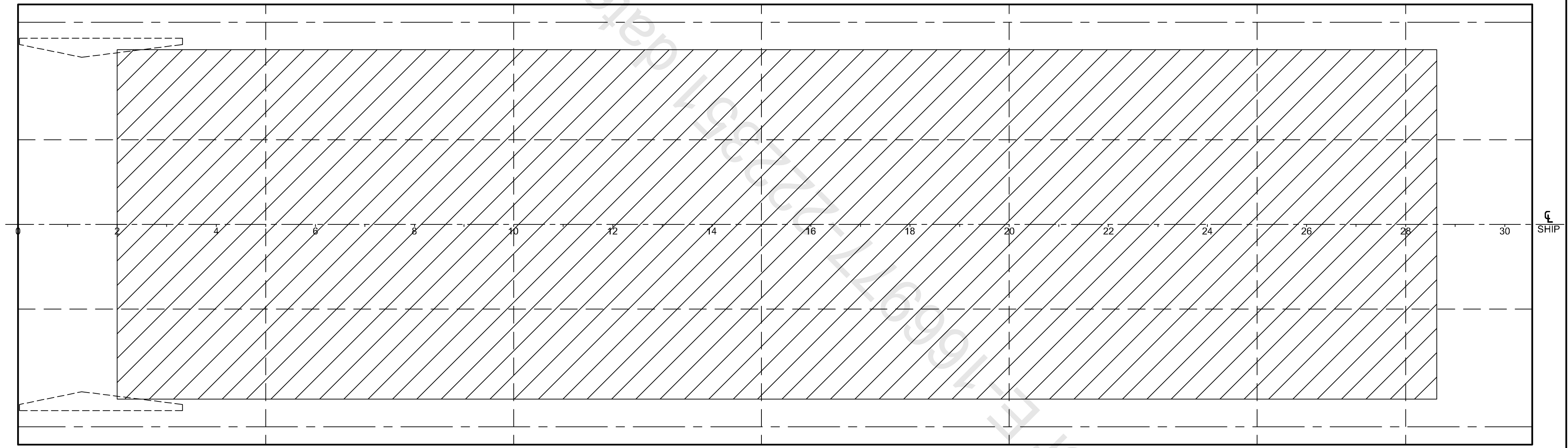


No.	Criteria	Actual	Requirement	Pass/Fail
1	The area under the GZ curve from 0.0 degrees to 19.9 degrees (Max. GZ)	1.258 m.rad	0.08 m.rad	pass
2	The vanishing stability angle	77.6 degrees	20 degrees	pass
3	The heel due to applied heeling curve(11.126 T-m)	0.0 degrees	11.6 degrees	pass

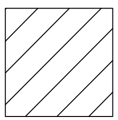
LOAD DEPARTURE CONDITION WITH CARGO



PROFILE



MAIN DECK PLAN



DECK CARGO

Condition 2: LOAD DEPARTURE CONDITION WITH CARGO

Item	Weight	LCG	LMom	VCG	VMom	TCG	FSM
DECK CARGO	1376.885	0.565	777.94	5.650	7779.40	0.000	0.000
Deadweight	**1376.885**	**0.565**	**777.94**	**5.650**	**7779.40**	**0.000**	**0.000**
Lightship	**388.635**	**1.332**	**517.66**	**3.000**	**1165.90**	**0.000**	**0.000**
Displacement	**1765.520**	**0.734**	**1295.60**	**5.067**	**8945.31**	**0.000**	**0.000**

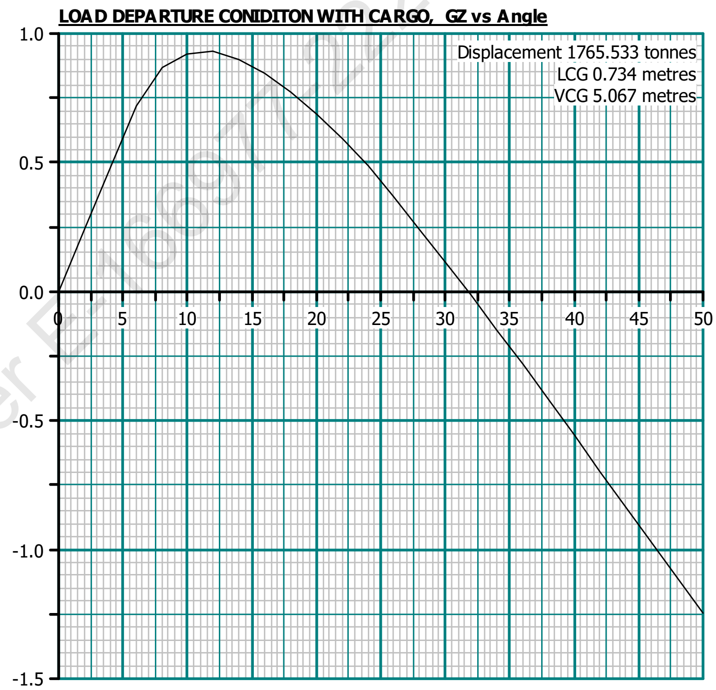
Draught	Aft	2.210 metres
	Mid	2.210 metres
	Fwd	2.210 metres

Trim Between Marks 0.000 metres by the stern

GM Solid 6.929 metres
GM Fluid 6.929 metres
Effective VCG 5.067 metres

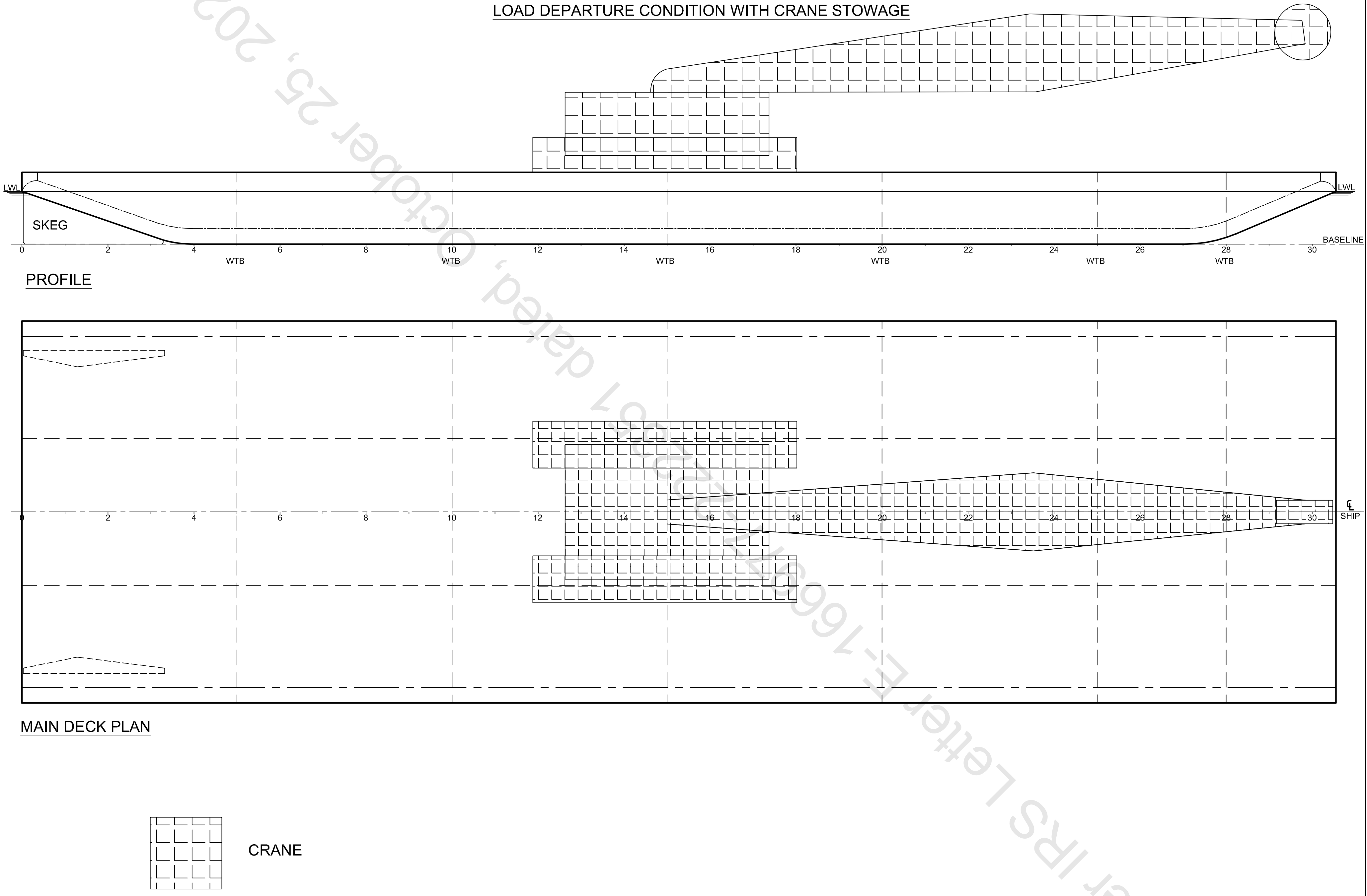
Moulded Displacement	1754.560 tonnes
Waterline at LCF referred to hull definition datum	2.200 metres
LCF referred to hull definition datum	0.454 metres
Heel Angle	0.00 degrees to starboard

Heel Angle degrees	Righting GZ metres	Lever KN metres	Waterline metres	Trim metres	VCB metres	GZ Curve Area metres.rad
0.0	0.000	0.000	2.200	0.000	1.152	0.000
10.0	0.923	1.803	2.244	-0.008	1.275	0.097
20.0	0.689	2.422	2.653	-0.029	1.390	0.246
30.0	0.118	2.651	3.128	-0.058	1.457	0.319
40.0	-0.557	2.700	3.539	-0.087	1.492	--
50.0	-1.245	2.636	3.874	-0.114	1.515	--



No.	Criteria	Actual	Requirement	Pass/Fail
1	The area under the GZ curve from 0.0 degrees to 11.5 degrees (Max. GZ)	0.121 m.rad	0.08 m.rad	pass
2	The vanishing stability angle	31.8 degrees	20 degrees	pass
3	The heel due to applied heeling curve (67.141 T-m)	0.3 degrees	3.1 degrees	pass

LOAD DEPARTURE CONDITION WITH CRANE STOWAGE



Condition 3: LOAD DEPARTURE CONIDITON WITH CRANE STOWAGE

Item	Weight	LCG	LMom	VCG	VMom	TCG	FSM
CRANE	231.250	0.500	115.63	4.930	1140.06	0.000	0.000
Deadweight	**231.250**	**0.500**	**115.63**	**4.930**	**1140.06**	**0.000**	**0.000**
Lightship	**388.635**	**1.332**	**517.66**	**3.000**	**1165.90**	**0.000**	**0.000**
Displacement	**619.885**	**1.022**	**633.29**	**3.720**	**2305.97**	**0.000**	**0.000**

Draught		
	Aft	0.839 metres
	Mid	0.849 metres
	Fwd	0.859 metres

Trim Between Marks 0.019 metres by the bow

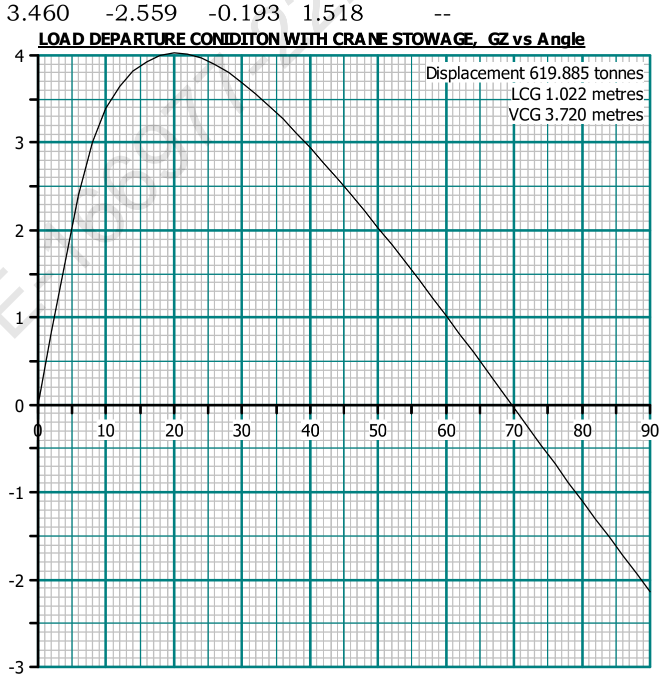
GM Solid 23.643 metres

GM Fluid 23.643 metres

Effective VCG 3.720 metres

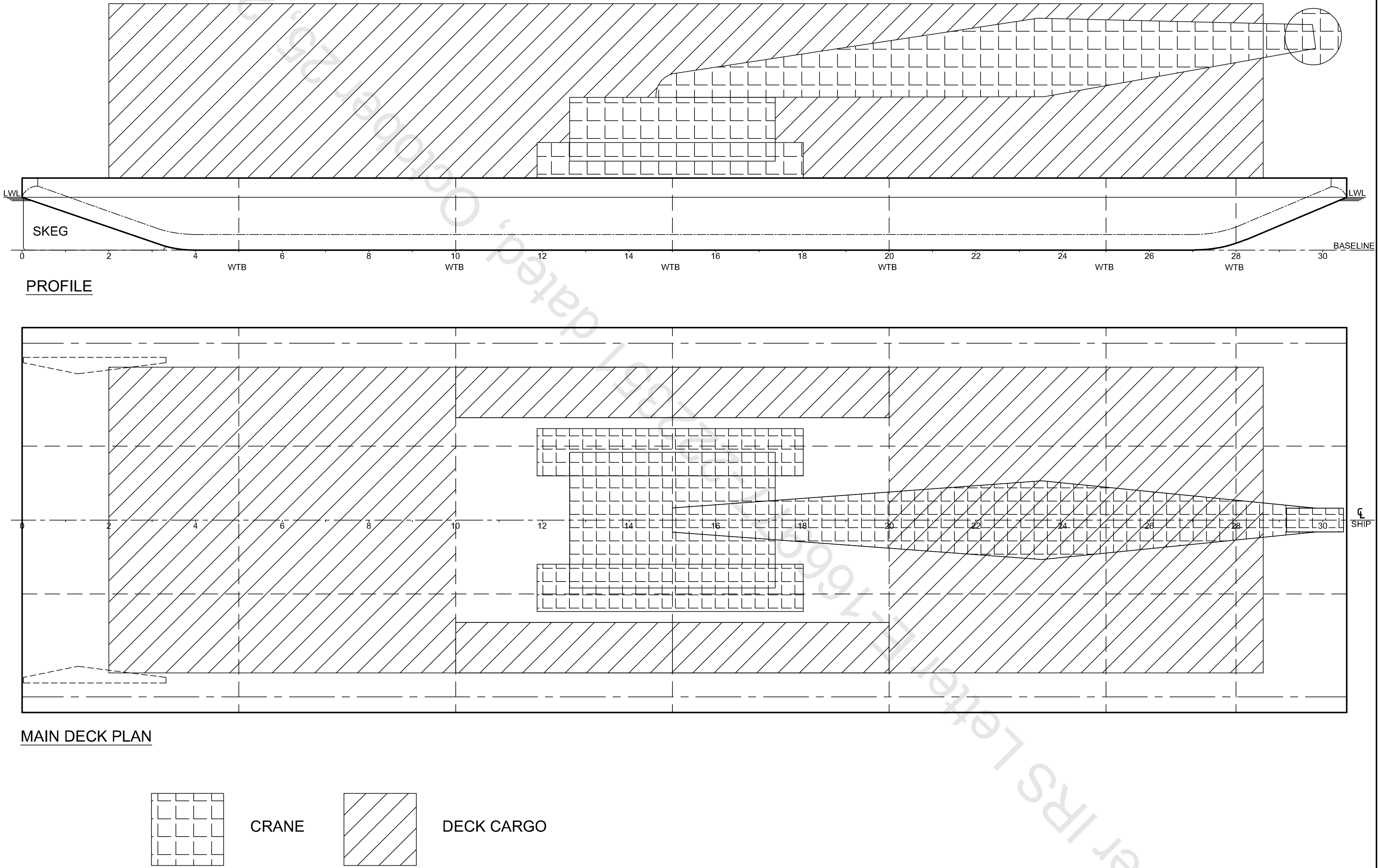
Moulded Displacement	611.551 tonnes
Waterline at LCF referred to hull definition datum	0.839 metres
LCF referred to hull definition datum	0.840 metres
Heel Angle	0.00 degrees to starboard

Heel Angle degrees	Righting GZ metres	Lever KN metres	Waterline metres	Trim metres	VCB metres	GZ Curve Area metres.rad
0.0	0.000	0.000	0.839	-0.019	0.435	0.000
10.0	3.390	4.036	0.725	-0.036	0.744	0.334
20.0	4.028	5.300	0.165	-0.049	1.071	1.001
30.0	3.687	5.547	-0.506	-0.083	1.262	1.684
40.0	2.947	5.338	-1.133	-0.116	1.363	2.266
50.0	2.032	4.882	-1.695	-0.147	1.429	2.703
60.0	1.022	4.243	-2.175	-0.173	1.478	2.970
70.0	-0.036	3.460	-2.559	-0.193	1.518	--



No.	Criteria	Actual	Requirement	Pass/Fail
1	The area under the GZ curve from 0.0 degrees to 20.5 degrees (Max. GZ)	1.033 m.rad	0.08 m.rad	pass
2	The vanishing stability angle	69.7 degrees	20 degrees	pass
3	The heel due to applied heeling curve (52.760 T-m)	0.2 degrees	8.6 degrees	pass

LOAD DEPARTURE CONDITION WITH CARGO & CRANE STOWAGE



Condition 4: LOAD DEPARTURE CONIDITON WITH CARGO & CRANE STOWAGE

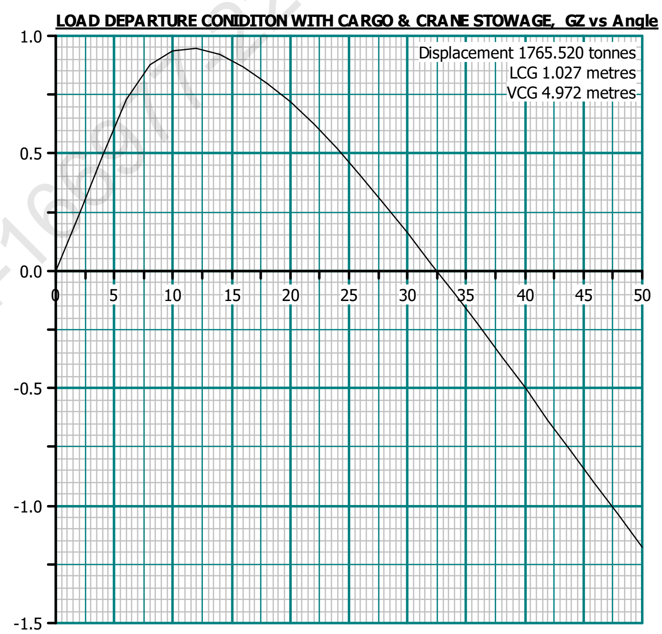
Item	Weight	LCG	LMom	VCG	VMom	TCG	FSM
DECK CARGO	1145.635	1.030	1180.00	5.650	6472.84	0.000	0.000
CRANE	231.250	0.500	115.63	4.930	1140.06	0.000	0.000
Deadweight	**1376.885**	**0.941**	**1295.63**	**5.529**	**7612.90**	**0.000**	**0.000**
Lightship	**388.635**	**1.332**	**517.66**	**3.000**	**1165.90**	**0.000**	**0.000**
Displacement	**1765.520**	**1.027**	**1813.29**	**4.972**	**8778.81**	**0.000**	**0.000**

Draught Aft 2.165 metres
Mid 2.208 metres
Fwd 2.252 metres

Trim Between Marks 0.087 metres by the bow

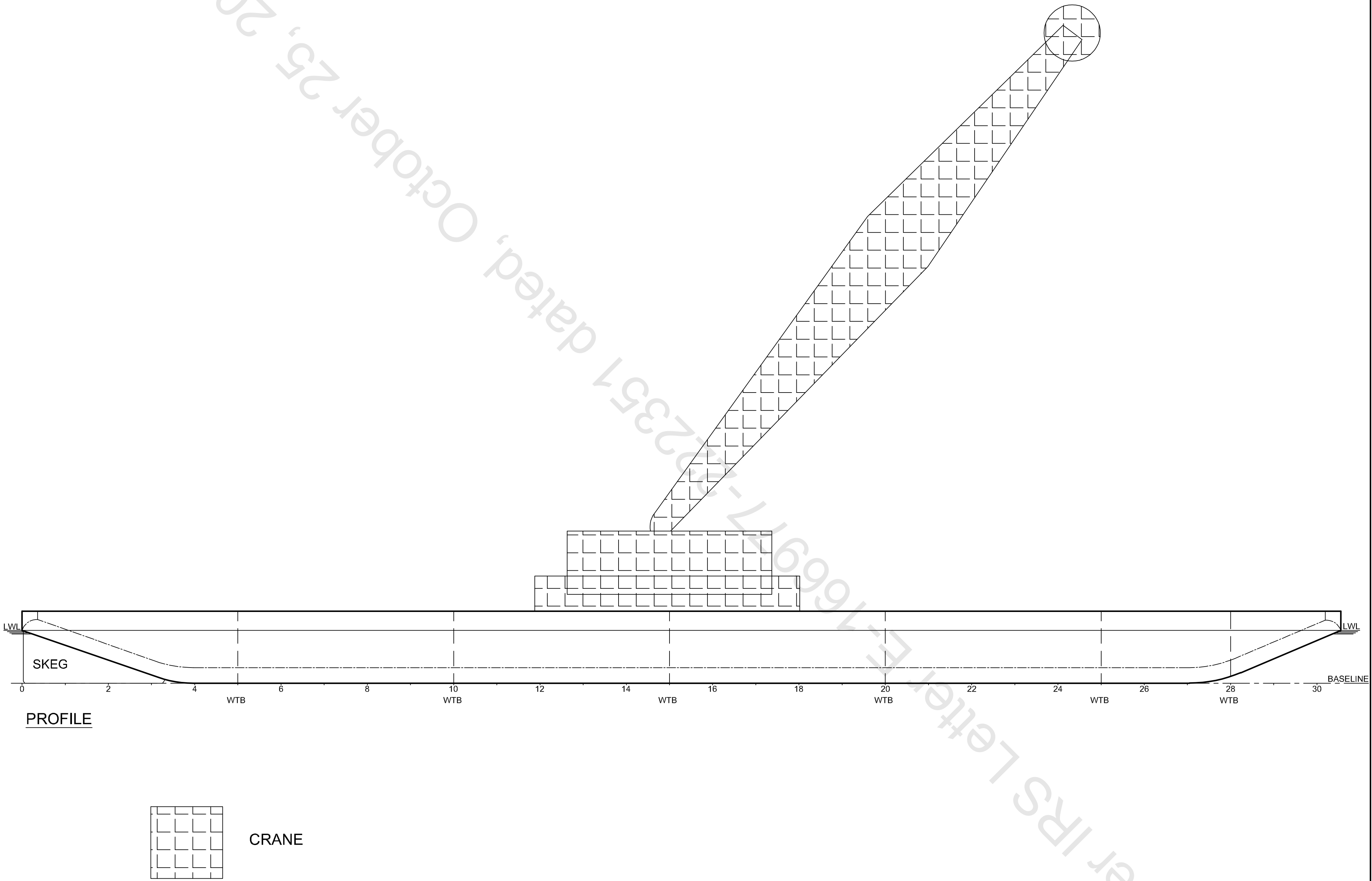
GM Solid 6.983 metres
GM Fluid 6.983 metres
Effective VCG 4.972 metres
Moulded Displacement 1754.623 tonnes
Waterline at LCF referred to hull definition datum 2.200 metres
LCF referred to hull definition datum 0.794 metres
Heel Angle 0.00 degrees to starboard

Heel Angle degrees	Righting GZ metres	Lever KN metres	Waterline metres	Trim metres	VCB metres	GZ Curve Area metres.rad
0.0	0.000	0.000	2.198	-0.087	1.153	0.000
10.0	0.937	1.800	2.242	-0.132	1.275	0.098
20.0	0.718	2.419	2.650	-0.230	1.390	0.251
30.0	0.161	2.647	3.122	-0.356	1.456	0.330
40.0	-0.500	2.696	3.532	-0.481	1.492	--
50.0	-1.176	2.633	3.864	-0.595	1.515	--



No.	Criteria	Actual	Requirement	Pass/Fail
1	The area under the GZ curve from 0.0 degrees to 11.7 degrees (Max. GZ)	0.126 m.rad	0.08 m.rad	pass
2	The vanishing stability angle	32.5 degrees	20 degrees	pass
3	The heel due to applied heeling curve (56.090T-m)	0.3 degrees	2.8 degrees	pass

CRANE OPERATION CONDITION - CRANE ONLY



Condition 5: CRANE OPERATION - CRANE ONLY

Item	Weight	LCG	LMom	VCG	VMom	TCG	FSM
CRANE	126.500	0.500	63.25	4.500	569.25	0.000	0.000
CRANE BOOM	21.100	0.500	10.55	45.900	968.49	0.000	0.000
COUNTER WEIGHT	83.700	0.500	41.85	5.730	479.60	0.000	0.000
CRANE CARGO	31.100	0.500	15.55	86.350	2685.49	0.000	0.000
Deadweight	**262.400**	**0.500**	**131.20**	**17.922**	**4702.83**	**0.000**	**0.000**
Lightship	**388.635**	**1.332**	**517.66**	**3.000**	**1165.90**	**0.000**	**0.000**
Displacement	**651.035**	**0.997**	**648.86**	**9.014**	**5868.73**	**0.000**	**0.000**

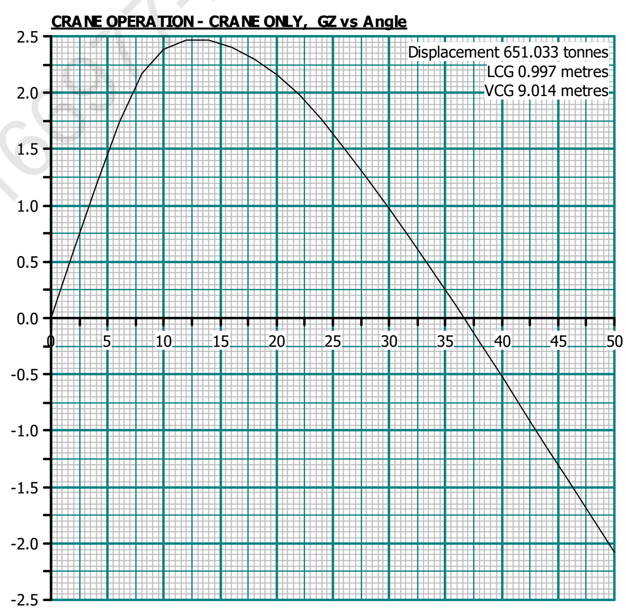
Draught Aft 0.881 metres
Mid 0.889 metres
Fwd 0.898 metres

Trim Between Marks 0.017 metres by the bow

GM Solid 17.098 metres
GM Fluid 17.098 metres
Effective VCG 9.014 metres

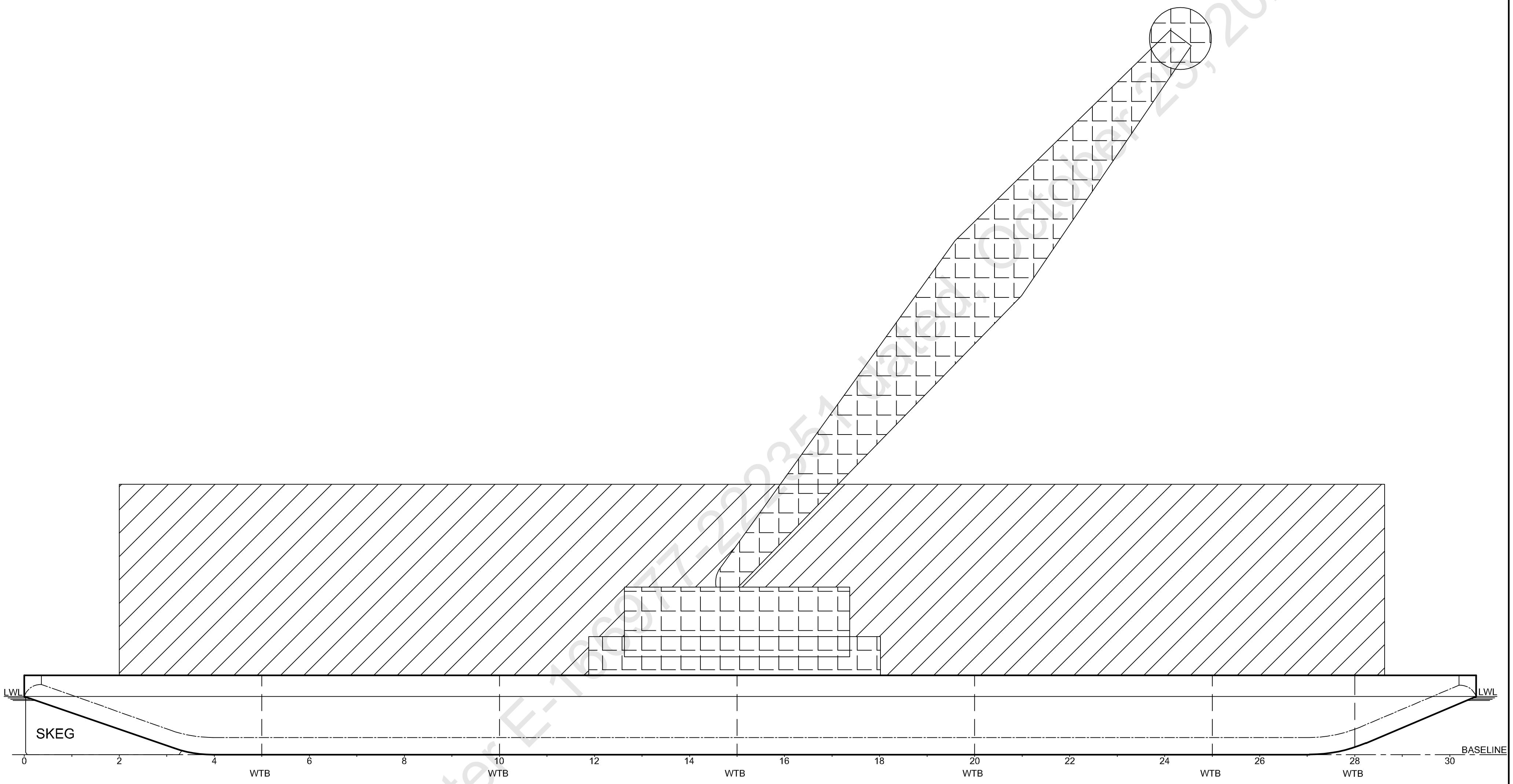
Moulded Displacement 642.714 tonnes
Waterline at LCF referred to hull definition datum 0.880 metres
LCF referred to hull definition datum 0.840 metres
Heel Angle 0.00 degrees to starboard

Heel Angle degrees	Righting GZ metres	Lever KN metres	Waterline metres	Trim metres	VCB metres	GZ Curve Area metres.rad
0.0	0.000	0.000	0.879	-0.017	0.455	0.000
10.0	2.386	3.951	0.774	-0.034	0.762	0.241
20.0	2.163	5.246	0.232	-0.050	1.094	0.658
30.0	0.973	5.480	-0.407	-0.084	1.276	0.939
40.0	-0.521	5.273	-1.006	-0.121	1.372	--
50.0	-2.080	4.825	-1.544	-0.154	1.435	--

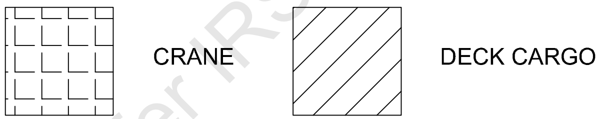


No.	Criteria	Actual	Requirement	Pass/Fail
1	The area under the GZ curve from 0.0 degrees to 12.8 degrees (Max. GZ)	0.360 m.rad	0.08 m.rad	pass
2	The vanishing stability angle	36.6 degrees	20 degrees	pass
3	The heel due to applied heeling curve (82.376 T-m)	0.4 degrees	8.4 degrees	pass

CRANE OPERATION CONDITION - CARGO & CRANE



PROFILE



Condition 6: CRANE OPERATION - CARGO & CRANE

Item	Weight	LCG	LMom	VCG	VMom	TCG	FSM
DECK CARGO	800.000	1.030	824.00	5.650	4520.00	0.000	0.000
CRANE	126.500	0.500	63.25	4.500	569.25	0.000	0.000
CRANE BOOM	21.100	0.500	10.55	45.900	968.49	0.000	0.000
COUNTER WEIGHT	83.700	0.500	41.85	5.730	479.60	0.000	0.000
CRANE CARGO	31.100	0.500	15.55	86.350	2685.49	0.000	0.000
Deadweight	**1062.400**	**0.899**	**955.20**	**8.681**	**9222.83**	**0.000**	**0.000**
Lightship	**388.635**	**1.332**	**517.66**	**3.000**	**1165.90**	**0.000**	**0.000**
Displacement	**1451.035**	**1.015**	**1472.86**	**7.160**	**10388.73**	**0.000**	**0.000**

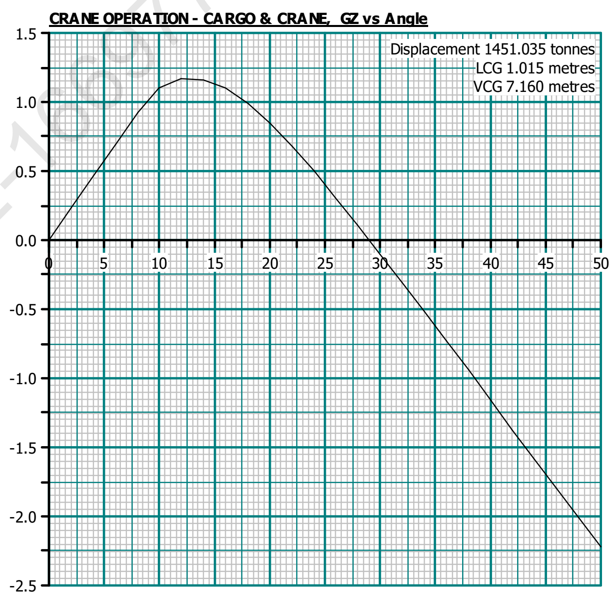
Draught Aft 1.820 metres
Mid 1.855 metres
Fwd 1.890 metres

Trim Between Marks 0.070 metres by the bow

GM Solid 6.811 metres
GM Fluid 6.811 metres
Effective VCG 7.160 metres

Moulded Displacement 1440.570 tonnes
Waterline at LCF referred to hull definition datum 1.846 metres
LCF referred to hull definition datum 0.736 metres
Heel Angle 0.00 degrees to starboard

Heel Angle degrees	Righting GZ metres	Lever KN metres	Waterline metres	Trim metres	VCB metres	GZ Curve Area metres.rad
0.0	0.000	0.000	1.845	-0.070	0.963	0.000
10.0	1.100	2.344	1.812	-0.086	1.149	0.100
20.0	0.843	3.292	1.965	-0.184	1.352	0.289
30.0	-0.098	3.482	2.123	-0.296	1.433	--
40.0	-1.159	3.443	2.247	-0.407	1.476	--
50.0	-2.221	3.264	2.333	-0.511	1.504	--



No.	Criteria	Actual	Requirement	Pass/Fail
1	The area under the GZ curve from 0.0 degrees to 12.7 degrees (Max. GZ)	0.154 m.rad	0.08 m.rad	pass
2	The vanishing stability angle	29.0 degrees	20 degrees	pass
3	The heel due to applied heeling curve (71.236T-m)	0.4 degrees	4.3 degrees	pass

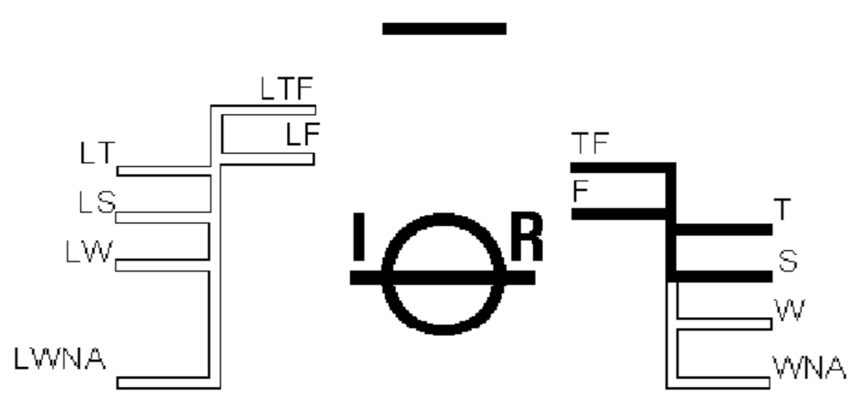
MEMORANDUM OF FREEBOARDS
25 Aug 2022

ARCADIA MINICA	Type B freeboard		
	Freeboard Length	:	52.80 m
	Summer WL Extreme Draught - loadline amidships	:	2.210 m

The upper edge of the deckline from which these freeboards are measured is 0 mm below top of * ~~Wood~~ / *steel main Deck *at side / *~~continued to side~~. (*Delete words not applicable).

Freeboards from Deck Line			**Seasonal, Fresh Water, or Timber allowances**	
Tropical Fresh Water	715	**mm.**	95	**mm. above summer**
Fresh Water	761	**mm.**	49	**mm. above summer**
Tropical	764	**mm.**	46	**mm. above summer**
Summer	810	**mm.**	**Upper edge of line through centre**	
Winter	Not	**mm.**	Assigned	**mm. below summer**
Winter North Atlantic	Not	**mm.**	Assigned	**mm. below summer**
Timber Tropical fresh water	—	**mm.**	—	**mm. above timber summer**
Timber fresh water	—	**mm.**	—	**mm. above timber summer**
Timber Tropical	—	**mm.**	—	**mm. above timber summer**
Timber Summer	—	**mm.**	—	**mm. above centre of ring**
Timber Winter	—	**mm.**	—	**mm. below timber summer**
Timber Winter North Atlantic		**mm.**		**mm. below timber summer**

VALID ONLY FOR UNMANNED OPERATION WHILE UNDER TOW ALONG THE INDIAN COAST, DURING THE COURSE OF WHICH THE VESSEL DOES NOT GO MORE THAN 20 NAUTICAL MILES FROM THE NEAREST LAND, AND MAY CROSS GULFS OR SIMILAR FEATURES RECOGNISED BY THE ADMINISTRATION.

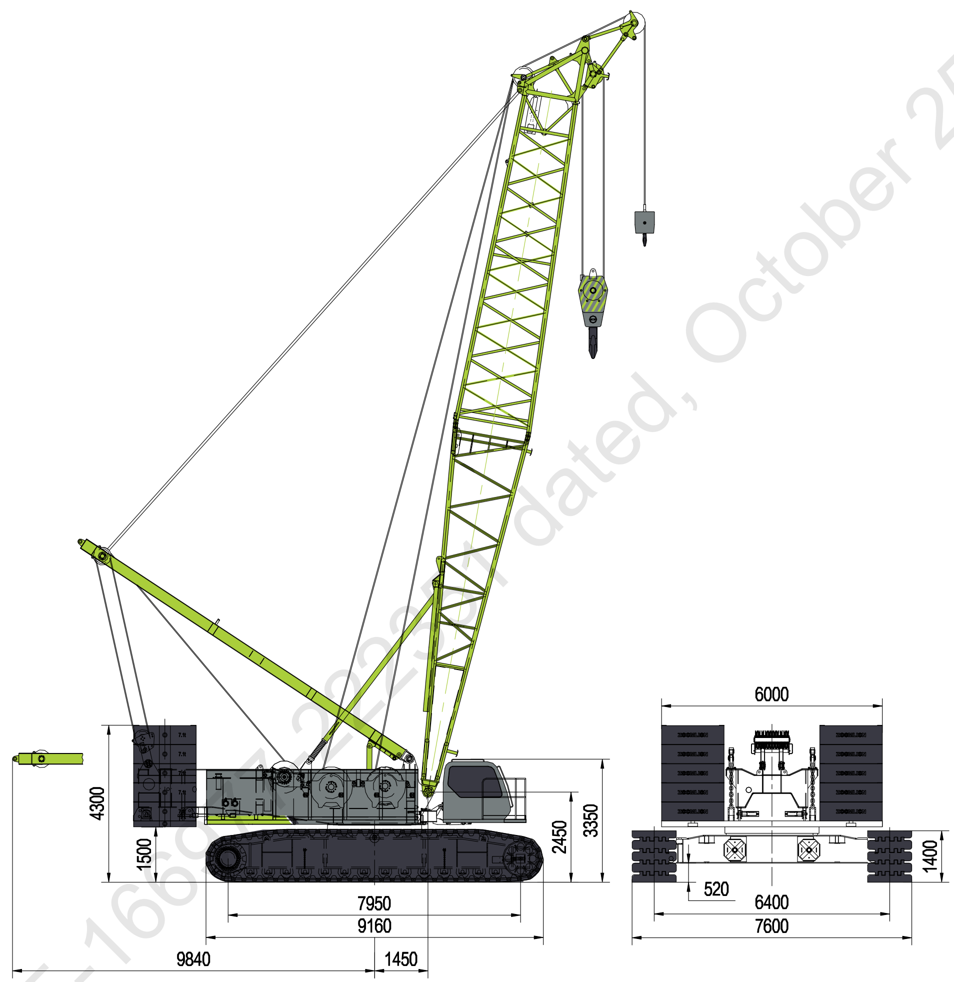


200 mm aft of fr.16

Form No. 4111 (07-83)

DIMENSIONS

Basic machine with undercarriage



Operating weight

The operating weight includes the basic machine with crawlers,2 main winches including wire ropes(300m and 480m) and 20m main boom, consisting of A-frame, boom foot (10m),boom head(10m),83.7t basic counterweight,32t carbody counterweight,and 260t hook block.

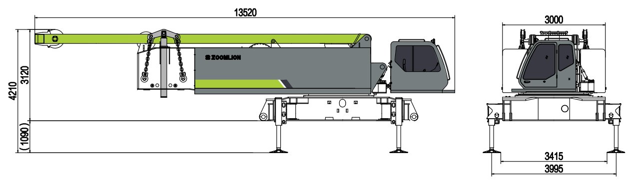
Total weight--approx.220t

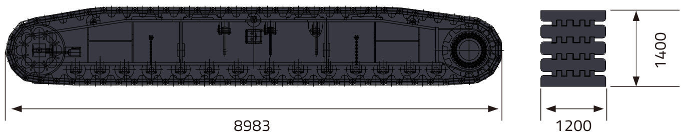
Equipment

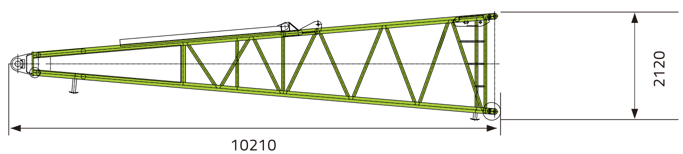
Main boom max. length---83m
High reach--95m
Luffing jib max. length--60m
Max. combination --main boom 62m
---luffing jib 60m
Fixed jib ---12-30m
Auxiliary jib 25t

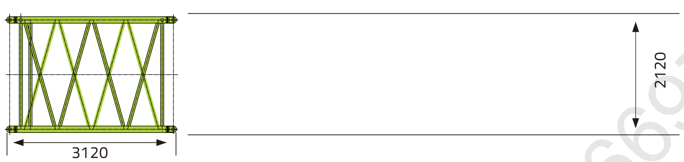
TRANSPORT DIMENSIONS AND WEIGHTS

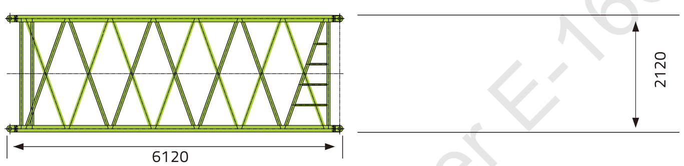
Basic machine and main boom

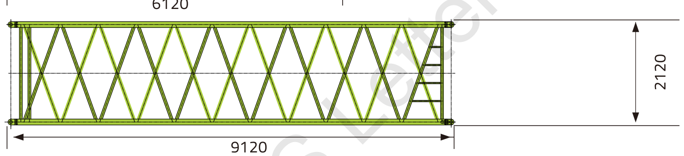


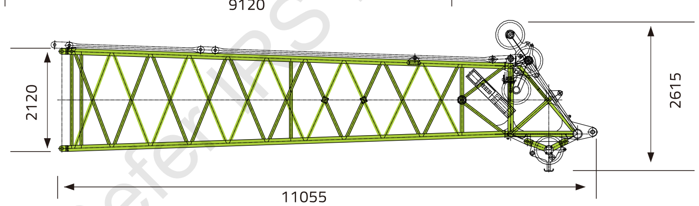












Basic machine	
Width	3000mm
Weight A	32000kg
(Without hoisting winch, main boom tilting-back support, A-frame and main boom derricking winch)	
Weight B	37500kg
(Without hoisting winch and main boom tilting-back support)	
Weight C	45500kg
(Including hoisting winch, main boom tilting-back support, A-frame and main boom derricking winch)	

Crawler	**2x**
Track pads	1200mm
Width	1200mm
Weight	24500kg

Boom foot	**1x**
Width	2365mm
Weight	3300kg

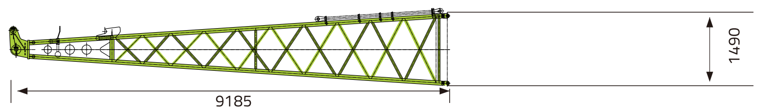
3m boom section	**1x**
Width	2320mm
Weight	850kg

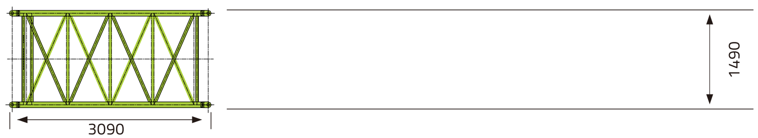
6m boom section	**1x**
Width	2320mm
Weight	1400kg

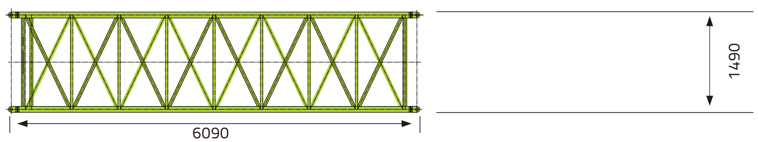
9m boom section	**6x**
Width	2320mm
Weight	2000kg

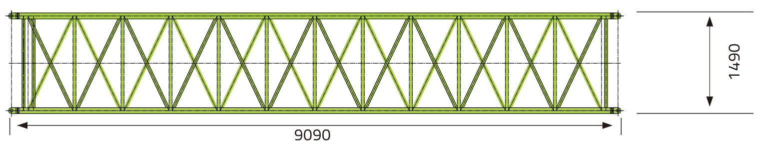
Boom head	**1x**
Width	2320mm
Weight	3500kg

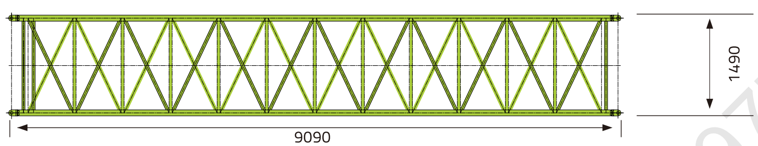
TRANSPORT DIMENSIONS AND WEIGHTS
Luffing jib

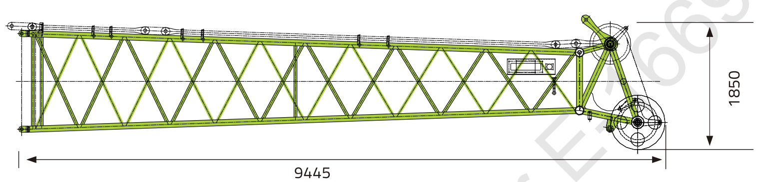


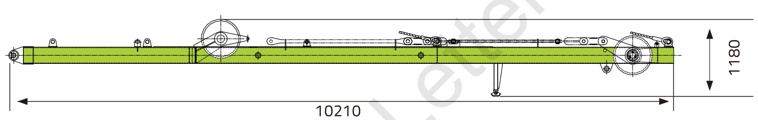


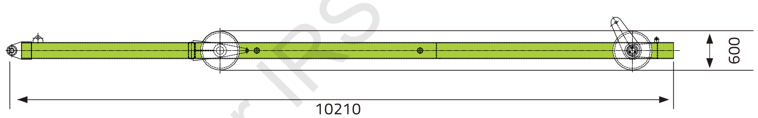












Luffing jib foot	**1x**
Width	1690mm
Weight	1100kg

3m luffing jib section	**1x**
Width	1690mm
Weight	370kg

6m luffing jib section	**2x**
Width	1690mm
Weight	660t

9m luffing jib section	**1x**
Width	1690mm
Weight	970kg

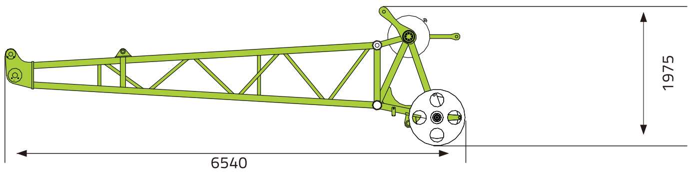
9m luffing jib section B	**2x**
Width	1690mm
Weight	830kg

Luffing jib head	**1x**
Width	1690mm
Weight	1100kg

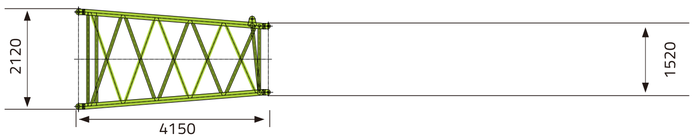
WA-frame 2	**1x**
Width	1750mm
Weight	1750kg
WA-frame 1 (width: 1385 mm) – 1.63 t × 1	

WA-frame 1	**1x**
Width	1385mm
Weight	1630kg

TRANSPORT DIMENSIONS AND WEIGHTS

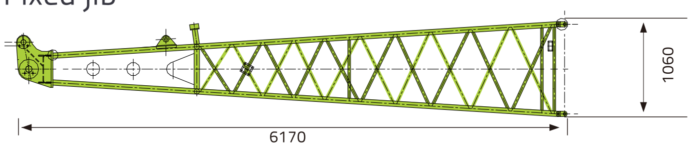


Heavy-duty fixed jib	1x
Width	1565mm
Weight	1000kg

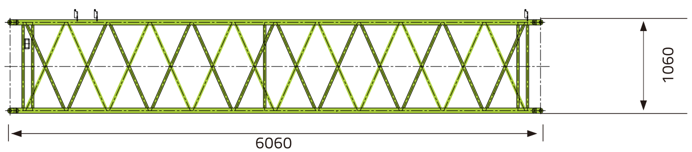


4m L-boom section tapered	1x
Width	2320mm
Weight	1100kg

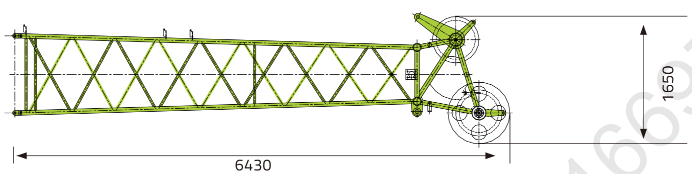
Fixed jib



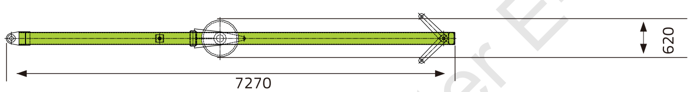
Fixed jib food	1x
Width	1540mm
Weight	500 kg



6m fixed jib section	3x
Width	1270mm
Weight	360 kg

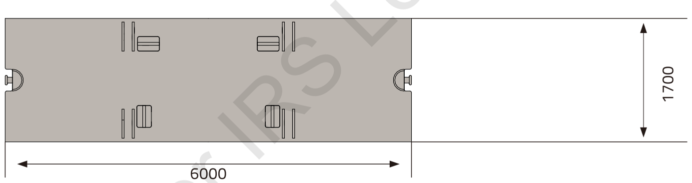


Fixed jib head	1x
Width	1270mm
Weight	640kg
FA-frame (width: 1500 mm) – 0.75 t × 1	



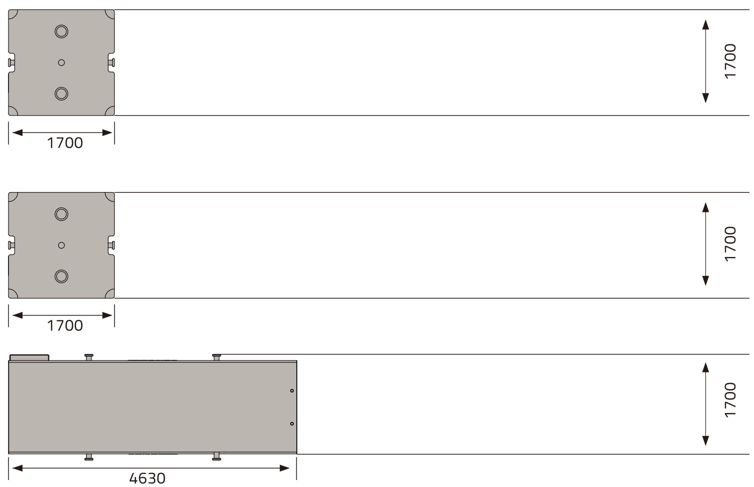
FA-frame	1x
Width	1500mm
Weight	750 kg

Counterweight

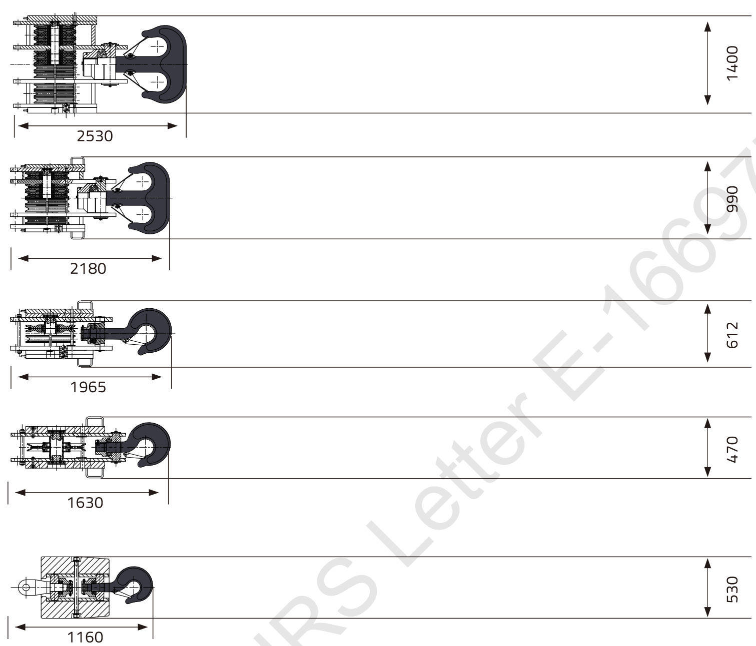


Counterweight base plate	1x
Thickness	160mm
Weight	12700kg
Counterweight plate (thickness: 510 mm) – 7.1 t × 10	

TRANSPORT DIMENSIONS AND WEIGHTS



Hooks



Counterweight	**10x**
Thickness	510mm
Weight	710kg
Counterweight plate (thickness: 300 mm, TBM operating mode for options) – 4.0 t × 2	

Counterweight for TBM operating mode	**2x**
Thickness	300mm
Weight	4000kg
Central counterweight (thickness: 650 mm) – 16 t × 2	

Carbody counterweight	**2x**
Thickness	650mm
Weight	16000kg

260t hook blocks-10 sheaves	**1x**
Weight	4230kg

160 t /100 t hook blocks-6 sheaves	**1x**
Weight	2780kg

50t hook blocks-2 sheaves	**1x**
Weight	1700kg
30 t load hook – 1.08 t × 1	

30t hook blocks-1sheave	**1x**
Weight	1080kg

16t single hook	**1x**
Weight	530kg

BOOM COMBINATION

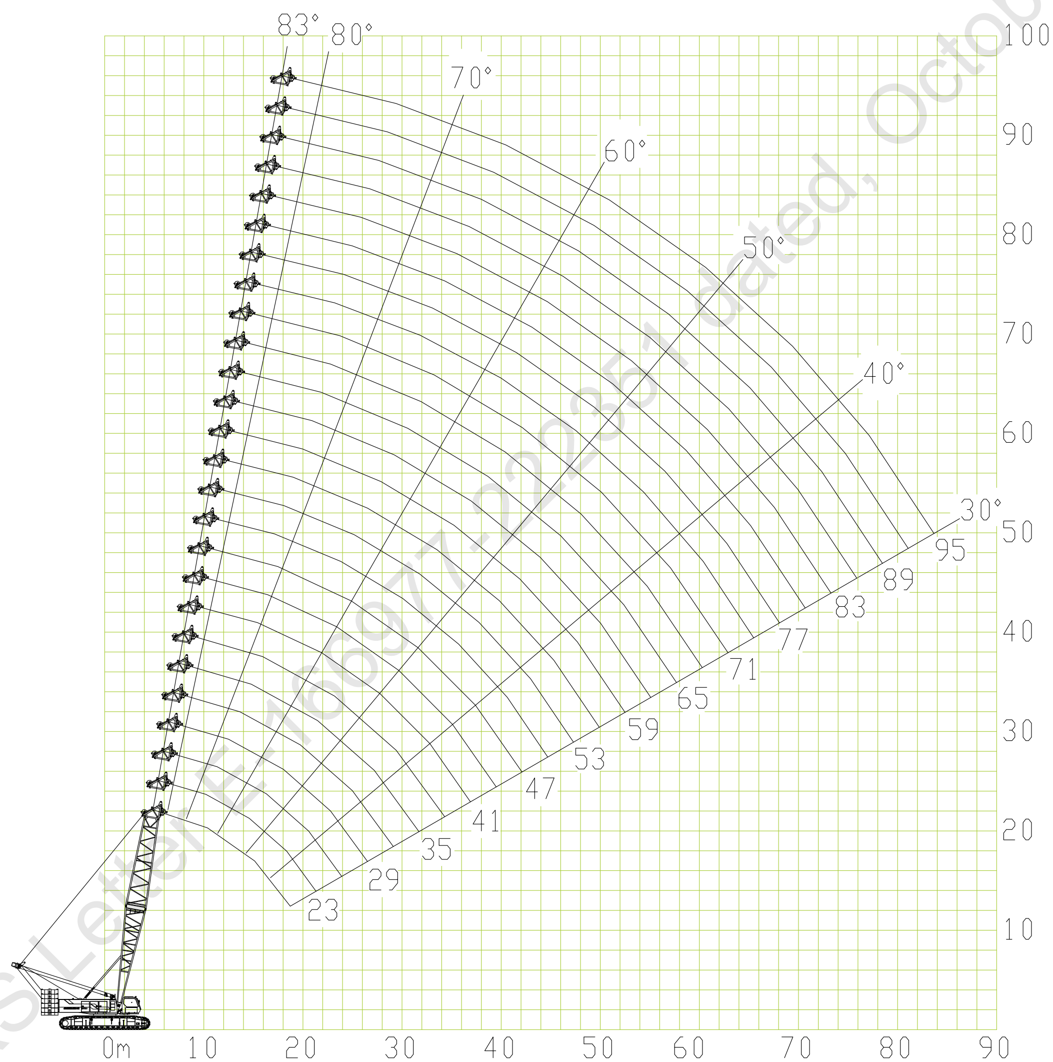
Descriptions fo Boom Assembly Codes		
Code	Type	Operation mode parameters
S	Heavy duty boom	20-83m
SL	Light dusy boom	86-95m
SW	Luffing jib	Main boom:23-62m jib:21-60m

Descriptions fo Boom Assembly Codes		
Code	Type	Operation mode parameters
SF	Fixed jib	Main boom:29-77m jib:12-30m
SFV	Heavy fixed jib	Main boom:41-77m jib:6m

LIFTING PERFORMANCE

Lifting performance on S/SL boom

Lifting height on S/SL boom



LIFTING PERFORMANCE

S-boom length (m)	Lifting capacity on S boom										
	20	23	26	29	32	35	38	41	44	47	50
Radius											
5	260										
6	227	200.8	193.1	184							
7	190.7	190.5	189	174	163	161					
8	175	166.6	164.5	162.5	158.4	149	149	135	118		
9	134.6	134.5	134.5	134.3	134.3	134.2	134.1	124	112	112	
10	132	130	128	125.9	123.9	121.9	121.9	119.8	110	110	102
12	102.9	102.7	102.5	102.2	102	101.6	99.5	97.5	97.5	95.5	93.4
14	83	83	82.8	82.6	82.5	82.2	82.1	82	81.3	81.3	79.2
16	69.1	69	68.8	68.7	68.7	68.5	68.5	68.4	68.4	68.3	68.3
18	59.1	59	58.9	58.7	58.6	58.6	58.6	58.6	58.6	58.6	58.5
20		51.5	51.4	51.4	51.4	51.4	51.4	51.2	51.1	51	50.8
22			45.6	45.6	45.5	45.5	45.4	45.3	45.1	45	44.8
24			40.7	40.7	40.7	40.6	40.4	40.3	40.2	40.1	39.9
26				36.6	36.6	36.6	36.5	36.4	36.2	36.1	35.9
28					33.2	33.2	33.1	32.9	32.8	32.7	32.6
30						30.3	30.3	30.2	30	29.9	29.6
32						27.9	27.7	27.6	27.5	27.4	27.2
34							25.6	25.5	25.3	25.2	24.9
36								23.5	23.4	23	22.8
38									21.7	21.4	21.1
40									20	19.8	19.4
42										18.3	18
44											16.7

S-boom length (m)	Lifting capacity on S boom										
	53	56	59	62	65	68	71	74	77	80	83
Radius											
10	98	86									
12	93.4	85	83	80	70	68	60.9	55.4	50.3		
14	79.2	77.2	77.2	70	69	66.5	59.2	53.8	49	43.2	39.4
16	67	67	67	65	65	64	57.4	52.2	47.4	41.9	38
18	58.3	58.2	58	57.2	56.6	55.9	54.8	50.4	44.7	40.6	36.8
20	50.7	50.5	50.3	50.2	49.4	48.9	48.4	47.7	43.4	39.4	35.8
22	44.6	44.5	44.4	44.1	43.8	43.3	42.7	42.1	41.5	38.1	34.5
24	39.8	39.5	39.4	39.3	39.1	38.5	38	37.4	37	36.5	33.4
26	35.7	35.6	35.4	35.2	34.8	34.5	34.2	33.6	33.2	32.6	32.1
28	32.4	32.2	31.8	31.5	31.2	30.8	30.5	30.1	29.7	29.3	28.9
30	29.5	29.2	28.8	28.4	28.1	27.8	27.3	27	26.7	26.3	25.9
32	26.5	26.5	26.1	25.8	25.3	25	24.6	24.4	23.9	23.6	23.3
34	24.5	24.2	23.8	23.5	23	22.7	22.4	21.9	21.6	21.3	21
36	22.5	22.1	21.7	21.4	21.1	20.6	20.3	20	19.6	19.2	18.9
38	20.5	20.3	20	19.5	19.2	18.9	18.5	18.1	17.8	17.4	17.1
40	19	18.7	18.3	18	17.7	17.2	16.9	16.6	16.1	15.8	15.5
42	17.6	17.2	16.9	16.6	16.1	15.8	15.5	15	14.7	14.4	14
44	16.3	15.9	15.6	15.2	14.8	14.5	14.1	13.8	13.4	13.1	13
46	15.1	14.8	14.4	14	13.7	13.3	12.9	12.6	12.4	12.2	12
48		13.7	13.4	12.9	12.6	12.3	11.8	11.7	11.5	11.3	11
50		12.7	12.4	11.9	11.6	11.3	11	10.9	10.7	10.5	10
52			11.4	11.1	10.7	10.3	10.3	10.1	9.9	9.7	9.3
54				10.2	9.9	9.5	9.1	9.4	9.2	8.9	8.5
56					9.1	8.9	8.8	8.7	8.5	8.2	8.2
58					8.4	8	8.3	8.1	7.9	7.6	7.3
60						7.4	7.7	7.5	7.3	7	6.6
62							7.2	7	6.8	6.5	6.2
64								6.5	6.3	6	5.6
66								5.7	5.8	5.5	5.1
68									5.2	5.1	4.6
70										4.7	4.5
72											3.8

NOTE:

When tip boom is used, the lifting capacity on tip boom is the same as that on heavy duty boom at the same radius, but its max. lifting capacity should not exceed 25 t.

ARCADIA MINICA

YARD NO: 113

LIGHTSHIP REPORT

DRG NO: EEPL/113/19

OWNER

ARVIND & CO. SHIPPING AGENCIES PVT. LTD.,

BUILDER

ESSFOUR ENGINEERING PVT. LTD,
SURAT

PROGRAM USED
HYDROSTATICS, LOADING & DAMAGE PROGRAM SUITE
VERSION 30.01.12 FV 18.13.12.4

DEVELOPED BY
WOLFSON UNIT M.T.I.A
SOUTHAMPTON UNIVERSITY
ENGLAND

APPROVED
DATE: 23 Sep 2022
INDIAN REGISTER OF SHIPPING
MUMBAI

SEE LETTER E-166235-221864

ARCHETYPE

112, 1st Floor, Anand Tower II,
Airport Road, Chicalim,
Goa – 403711

TOTAL 04 SHEETS

LIGHTSHIP DATA

Vessel	: ARCADIA MINICA
Owner's Name and Address	: Arvind & Co. Shipping Agencies Pvt Ltd,
Place	: Essfour Engineering Pvt Ltd Rassaim, Goa.
Yard No.	: 113
Date	: 16th Sept. 2022
Total no. of persons on board	: Nil
Weather	: Calm
Wind	: Calm
Restraining Line	: All Slack
State of the Vessel	: Complete in all respect & All bilges Dry
Specific gravity	: 1.005
Temperature	: 30 deg C
Draft witnessed by	: 1) Mr. Anil Kumar (Goa) Builders Rep. 2) Mr. Pratamesh Chari, IRS (Goa) Rep.
Main Dimension	: L – 55.00m x B – 16.00m x D – 3.00m

DRAFT AT TIME OF LIGHTSHIP SURVEY

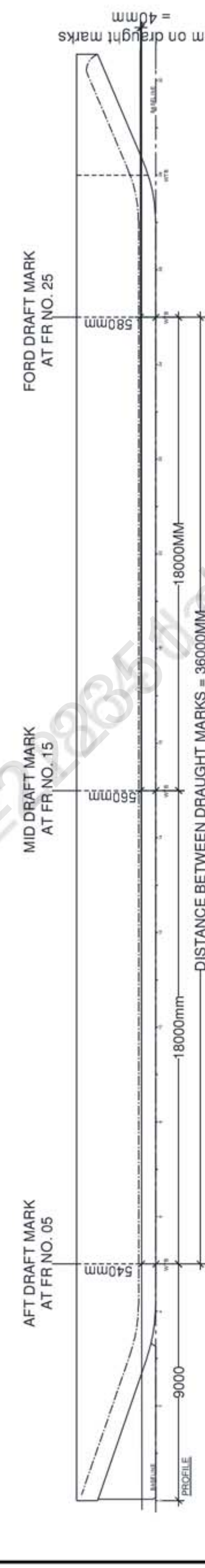
		P	S	Mean
Draft Aft (Fr no. 05)	:	0.540 m	0.540 m	0.540 m
Draft Ford (Fr. No. 25)	:	0.580 m	0.580 m	0.580 m
Draft Midship (Fr.No 15)	:			0.560 m
Trim by ford on Draft Mark	:			-0.040 m

Soundings Of Tanks

ALL TANKS EMPTY

Sign Convention: Longitudinal Position (LCG, LCB, LCF etc) are measured from midship (Fr. No 15) 27.0 mtrs forward of (Fr No. 0). Aft of Midship is considered –ve and forward +ve. All vertical measurements (VCG, VCB etc) are measured from baseline. Trim Aft + ve Trim Ford - ve

INCLINED WATER LINE DETAIL



INCLINED WATERLINE HYDROSTATICS REPORT

ACARDIA MINICA

Ship Particulars

Waterline Length	55.000 metres
Waterline Beam	16.000 metres
Top of Keel	0.000 metres
CP and CM referred to Section	55M Pontoon, FR NO. 15: X=0 metres
Specific Gravity of Water	1.0050
Mean Shell Thickness	0.0100 metres
Longitudinal Datum	FR NO. 15 (27MTRS FROM FR NO. 0)
Vertical Datum	BASELINE
Trim Length	36.000 metres

Draught Marks	**Name**	**X metres**	**Z metres**
Aft Marks	FR NO. 05	-18.000	-0.010
Mid Marks	Midships	0.000	-0.010
Fwd Marks	FR NO. 25	18.000	-0.010

Trim Between Marks 0.040 metres by the bow

Draught at Mid Marks metres	**Full Displacement tonnes**	**LCB metres**	**LCF metres**	**Moulded VCB metres**	**Immersion tonnes/cm**	**Moulded KMT metres**	**Moulded KML metres**	**MCT tonnes-metres/cm**
0.560	388.635	1.332	0.862	0.285	7.500	40.888	366.209	38.744

LIGHTSHIP DATA OF VESSEL YARD NO.113

SPECIFIC GRAVITY -		1.005
DISPLACEMENT	–	388.635 tonnes
LCG w.r.t Fr No. 15	–	1.332 mtrs
LCG w.r.t Transom	–	28.332 mtrs
VCG w.r.t. baseline	–	3.000 mtrs (at deck level)